ELECTRONIC SERVICE MANUAL

SVM 20i Series 32*70 I

Material Number of Documentation SM-07.05.02

THERMAL TRANSFER OVERPRINTER

SAVEMA Marking and Coding Machines

(ENG) English

Version: 27.01.25 (V7)

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1 PRINTER SETTINGS

Printer settings provides to;

- Change some values (delay, contrast and print position) while printer was printing.
- Edit system date and time.
- Adjust parameters.
- Load template to system.
- Choose and edit template.
- Change system language.
- Control sensors, valves, motors and fuses.
- Change IP host number.
- Entering the desired print quantity

1.1 MAIN MENU

When the machine started, the following screen will appear. The functions on the main menu are explained one by one.

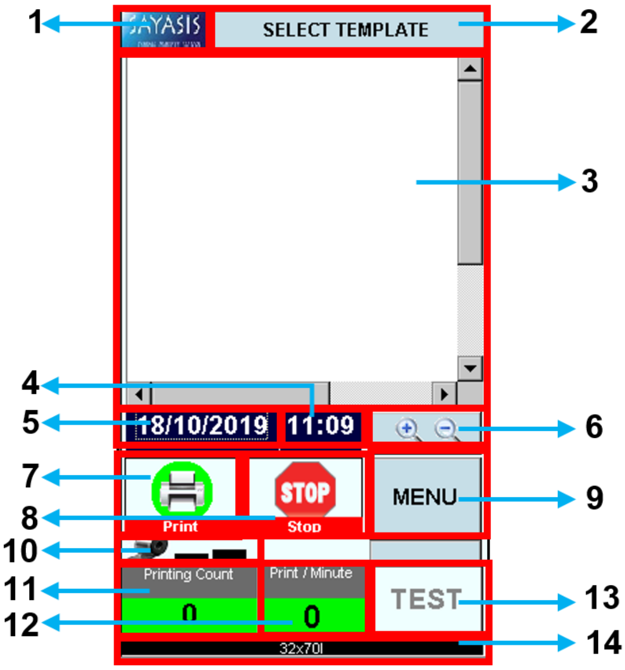


Figure 1. Main Menu

No	Designation
1	SAYASIS LOGO
2	SELECT TEMPLATE
3	TEMPLATE SCREEN
4	SYSTEM TIME
5	SYSTEM DATE
6	ZOOM
7	PRINT START
8	PRINT STOP
9	MENU
10	REMAINING RIBBON
11	PRINTING COUNT
12	PRINT/MINUTE
13	TEST
14	MACHINE TYPE

Table 1. Main Menu Functions

1. SAYASIS LOGO: This logo provides to open hidden settings window. When the clicked on this logo, system wants to enter password that is system date. Password format is “dmyyyy”. *Example 1; password is “18102019” for this date “18/10/2019”. Example 2; password is “952019” for this date “09/05/2019”* ***(zero at the beginning of the month and day is not entered in the password).*** After password is entered correctly, the hidden settings screen is opened. The screen provides to adjust parameter that variables to the general work of the machine. Also it provides to learn Information’s of customer and machine ID. Authorization that variables of network adjusting is made from hidden settings screen. The logo is passive when printer started to print. If printer stops, the logo becomes active.

2. SELECT TEMPLATE: If the module doesn't have any template, the template must be installed from USB to internal memory. The Print and Stop functions will not be active until template has been selected. When the first template is loaded to system, Print function is active. If desired to change template after printer started, no need to stop the printer. This function (select template) is active every time (printer working state is on or off) after first working.

3. TEMPLATE SCREEN: This area shows chosen template. The template can't change via this screen. System will not response when was touched on this area. On this area, template can be scrolled to right or left via horizontal scroll bar and to up or down via vertical scroll bar. The scrolling does not affect printing.

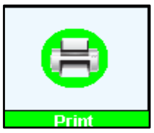
4. SYSTEM TIME: This area shows the current system time. When the clicked to system time area, the time can be increased or decreased respectively "↑"and "↓" symbols. This function is active every time so the system time can be changed every time printer works or not.

5. SYSTEM DATE: This area shows the current system date. The correct password for hidden setting is this date. If today is 06/05/2019 but system date is 02/01/2008, the correct password is 212008 not 652019. When the clicked to system date area, the date can be increased or decreased respectively "↑"and "↓" symbols. This function is active every time so the system date can be changed every time printer works or not.

6. ZOOM: This function provides to zoom in or zoom out to selected template. Unless the template is selected, zoom feature is not active. Zooming in or out not affect template that printed. Only shows template from different scale.

7. PRINT START: This button is used to start print. There is a condition to activate this function. The template should be chosen. The active and passive displays of the button are shown below.

Active Button 

Passive Button 

8. PRINT STOP: This button is used to stop print. There are 2 conditions to activate this function. First of all the template should be chosen and secondly printing should be started. The active and passive displays of the button are shown below.

Active Button 

Passive Button 

9. MENU: This button opens the screen that have functions provide to load template to system, choose and edit template and control sensors, valves, motors and fuses. The printer must be stopped to control sensors, valves, motors and fuses. The other situation, the printer doesn't have to stop. The menu button is active every time. Buttons that appear blue in this menu are active.

10. REMAINING RIBBON: This field shows how much ribbon is remain. Line length decreases as the ribbon decreases. This field is for illustration only. No value can be entered.

11. PRINTING COUNT: Shows the number of prints that printed since the machine was started. This field is for illustration only. No value can be entered. The value is reset in the area when the machine is restarted. Even if the template is changed, this value not reset.

12. PRINT/MINUTE: Shows the number of prints that printed in one minute. This field is for illustration only. No value can be entered. The value is reset in the area when the machine is restarted. Even if the template is changed, this value not reset.

13. TEST: The test button is used to print the template once. It will not be active until the template is selected.

14. MACHINE TYPE: This area shows whether it is pneumatic or magnetic and also shows whether the intermittent or continuous. This field is for illustration only. No value can be entered.

1.1.1 SELECT TEMPLATE SECTION

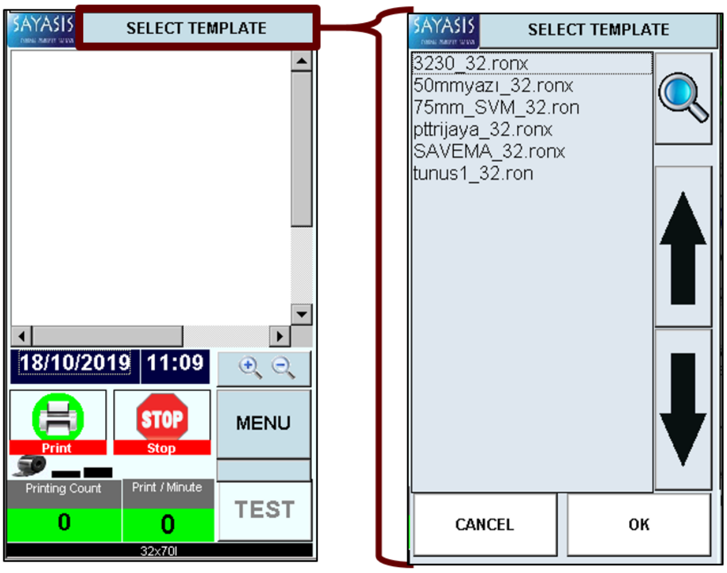


Figure 2. Template List Screen When Select Template Button Clicked

If the module doesn't have any template, the template must be installed from USB to internal memory. Choose desired template by using up or down arrow signs and press **OK** button. Unless press the button or if press **CANCEL** button, the chosen template is invalid. Press **OK** button before **CANCEL** button to be selected template is valid. After a five-second wait, selected template will appear on the screen.

1.1.2 SYSTEM DATE SECTION

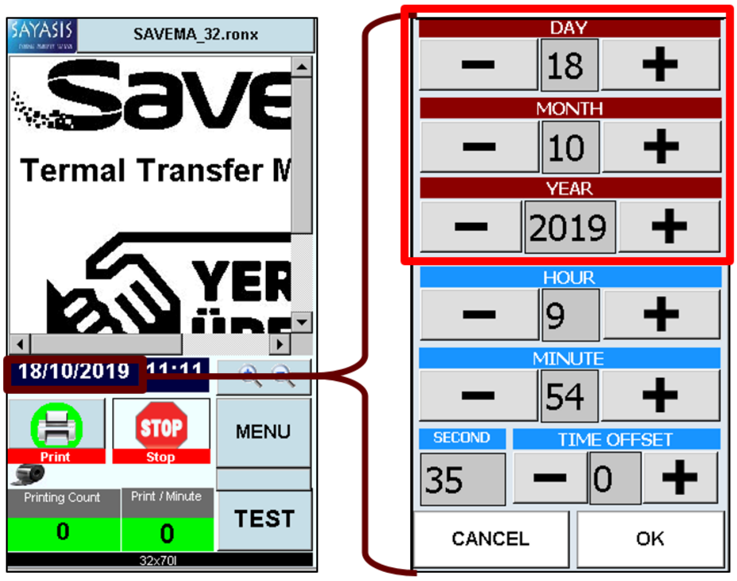


Figure 3. System Date Changing Screen When Date Button Clicked

System date can be changed with these settings. The date can be increased or decreased by using "+" and "-" buttons. System date is used for hidden settings and update templates. After the changes are completed, press **OK** button. Unless press the button or if press **CANCEL** button, the changes don't save. After pressing the **OK** button, if press **CANCEL** button, the changes are saved.

1.1.3 SYSTEM TIME SECTION

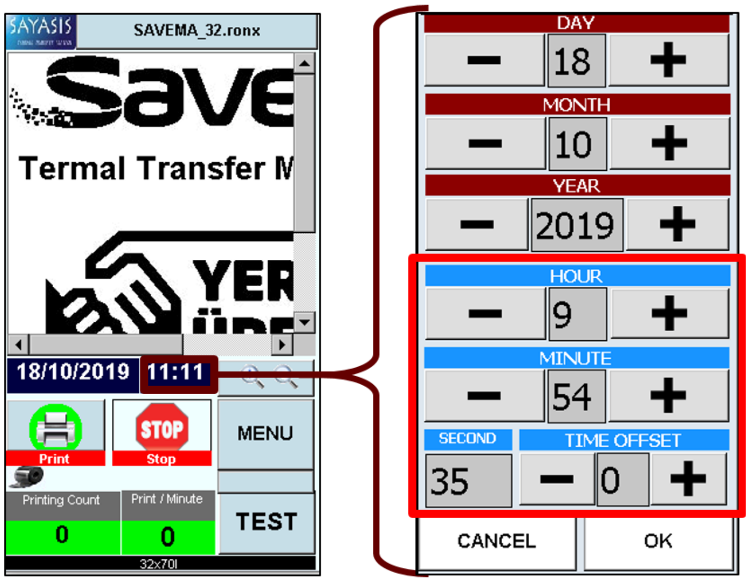


Figure 4. System Time Changing Screen When Time Button Clicked

System time can be changed with these settings. The date can be increased or decreased by using "+" and "-" buttons. System can be used time for update templates. After the changes are completed, press **OK** button. Unless press the button or if press **CANCEL** button, the changes don't save. After pressing the **OK** button, if press **CANCEL** button, the changes are saved.

1.1.4 MENU FUNCTIONS

Menu functions provide to load template to system, edit template and control sensors, valves, motors and fuses etc. Also some operations related to database and software settings can be done from this menu. The printer must be stopped to entering control section. The other situation, the printer doesn't have to stop. Buttons that appear blue are active.

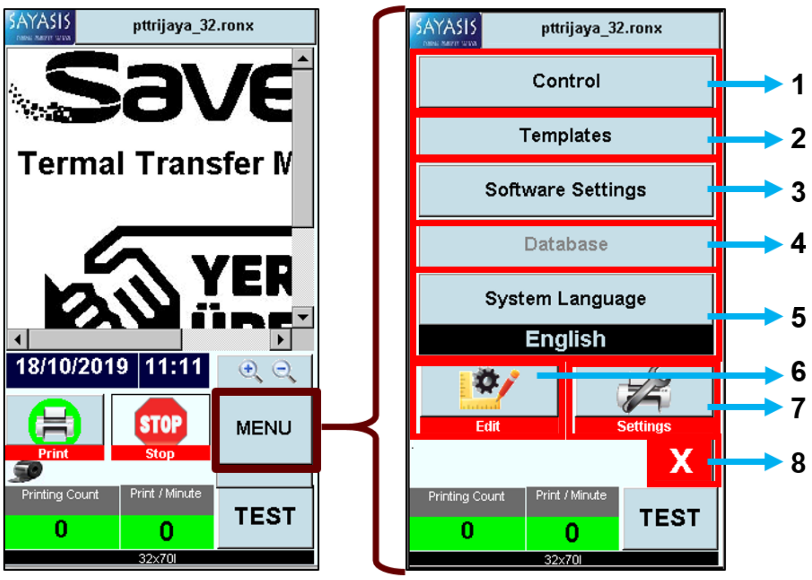


Figure 5. Menu Functions Screen

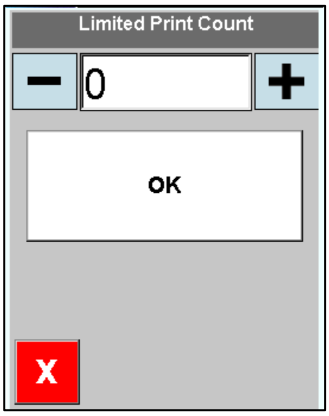
No	Designation
1	CONTROL
2	TEMPLATES
3	SOFTWARE SETTINGS
4	DATABASE
5	SYSTEM LANGUAGE
6	EDIT
7	SETTINGS
8	EXIT

Table 2. Menu Functions

1. CONTROL: Printer sensors, fuses, valves and motors can be controlled in this section. Also some signals that come to printer can be observed from the window. This function is not active while printer is printing. The printer must be stopped to enable this feature.

2. TEMPLATES: Templates can be saw in printer memory and USB stick memory. Templates can be inserted from USB stick to printer memory or vice versa. The button is active every time.

3. SOFTWARE SETTINGS: Software related operations can be made. See product user manual for detailed information.
4. DATABASE: Database related operations can be made. See product user manual for detailed information.
5. SYSTEM LANGUAGES: System languages can be changed via this section. The area is active every time.
6. EDIT: Some components of the template (barcode, text, monetary value etc.) can be edited via this property. The template must be selected to enable this feature.
7. SETTINGS: Settings window provides to change some values (delay, contrast and print position etc.) while printer was printing. Also ribbonsave mode and mirroring settings can be done on this menu. Unless the template is selected, settings button is not active.
8. EXIT: It is used to go back main menu.



LIMITED PRINT COUNT: Value that desired template quantity can be entered with the help of the **"+"** and "**–**" buttons on drop-down area. When clicked **"+"** button increases value while "**–**" button is decreases. For example; if 5000 is entered in this area, printer counts back from 5000. If the value equals to 0, printer stops. The value is reset in the area when the machine is restarted. If the template is changed, this value does not reset. There is a condition to activate this function. This condition is that template should be chosen.

First press the menu button to access this field. The "print / minute", "printing count" and "print / minute" fields are then pressed sequentially.

1.1.4.1 SETTINGS SECTION

Settings window provides to change some values (speed, delay, contrast, template shifting-direction and home position) while printer was printing. Ribbonsave mode and mirroring settings can be done at printing mode section on this menu. Unless the template is selected, settings button is not active.

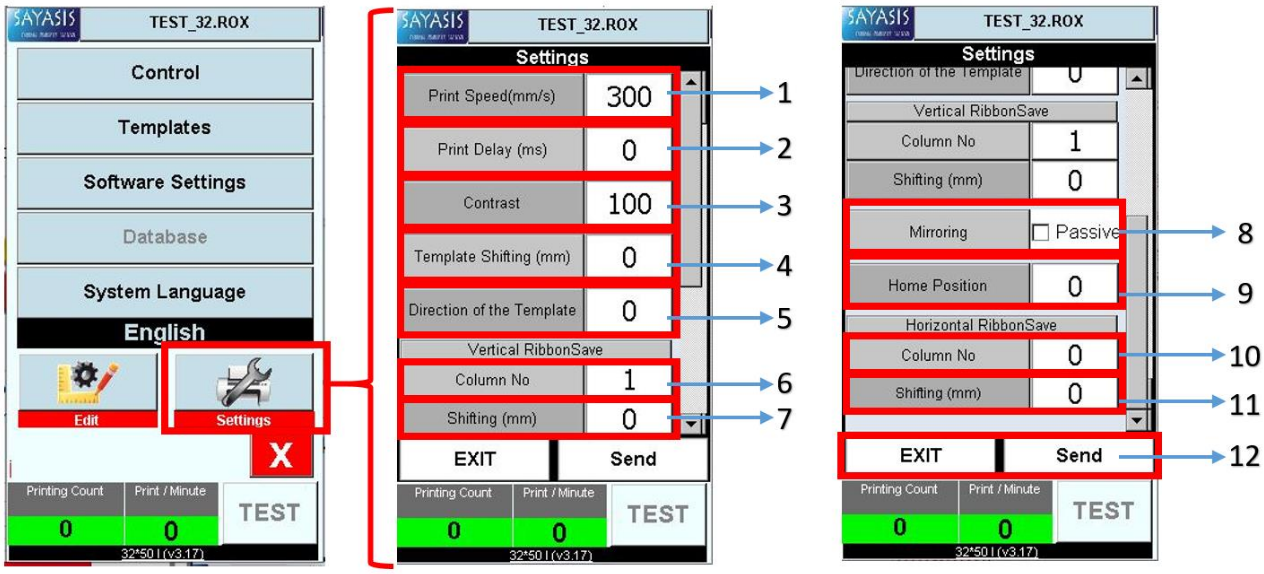


Figure 6. Settings Screen

No	Designation
1	PRINT SPEED
2	PRINT DELAY
3	CONTRAST
4	TEMPLATE SHIFTING
5	DIRECTION OF THE TEMPLATE
6	VERTICAL RIBBONSAVE/COLUMN NO
7	VERTICAL RIBBONSAVE/SHIFTING
8	MIRRORING
9	HOME POSITION
10	HORIZONTAL RIBBONSAVE/COLUMN NO
11	HORIZONTAL RIBBONSAVE/SHIFTING
12	OPERATING BUTTONS

Table 3. Settings Functions

1. PRINT SPEED: This function adjusts print length per second. The unit is mm/sec. That's mean; if 300 is entered to value, printer prints 300 mm in one second. "**+**" button increases the speed value and printed quantity in one second is increases. "**–**" button decreases the speed value and printed quantity in one second is decreases. The speed value cannot be negative. The maximum limit for speed value is 500. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

2. PRINT DELAY: Printer waits until the entered time in this box after contact signal came and does nothing. Printer prints after time-out. The unit is 10ms. That's mean; if 10 is entered to value, printer waits 100 ms. "**+**" button increases the delay value and print shifts more. "**–**" button decreases the delay value and print shifts less. The delay time cannot be negative. The maximum limit for delay time is 1000. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

3. CONTRAST: This function adjusts print darkness. The unit is none. "**+**" button increases the darkness value and print becomes darker. "**–**" button decreases the darkness value and print becomes lighter. The standard value is 100. The changeable range is 60-120 for contrast value. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

4. TEMPLATE SHIFTING: This function provides to determine the printing position along the thermal print head. The unit is mm. The shifting range is adjusted automatically by the printer according to the template. Positive values shift the template to the right and negative values shift the template to the left. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
   If the template in the picture below is shifted by 15mm, the template will look like the 2nd template below.

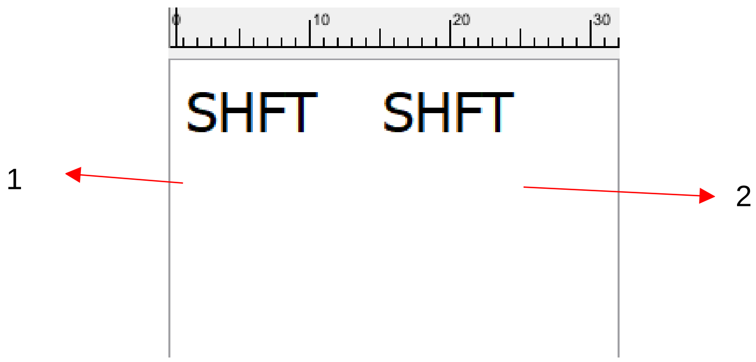


Figure 7 Template Shifting

No	Designation
1	MAIN TEMPLATE
2	SHIFTING TEMPLATE

Table 4 Template Shifting

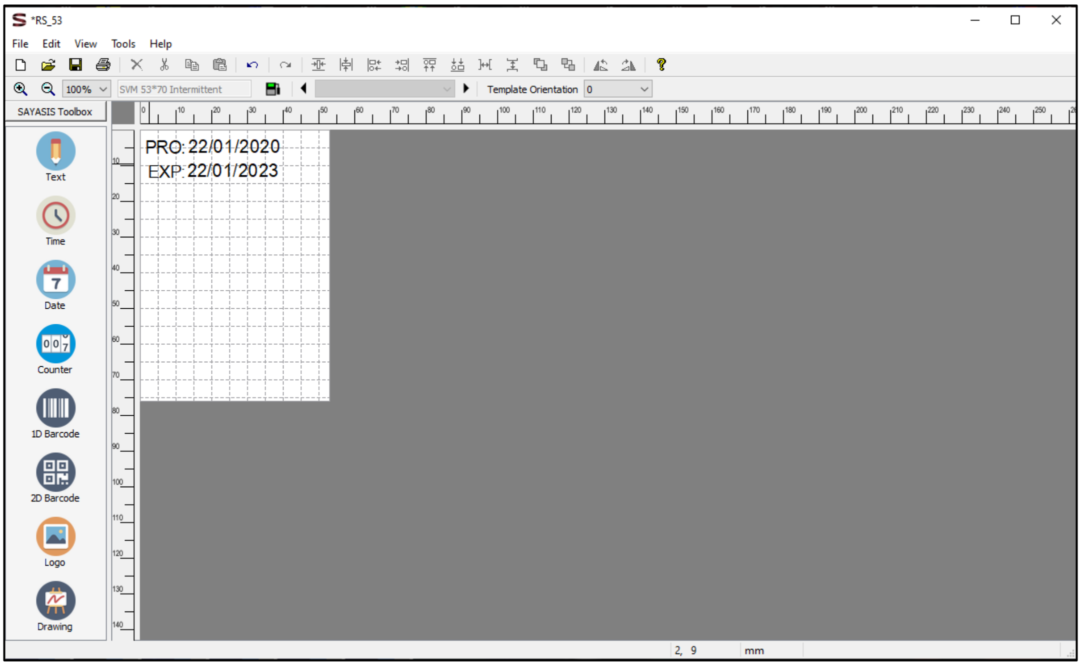
5. DIRECTION OF THE TEMPLATE: This feature provides to rotate template. By clicking down arrow sign, the desired rotation angle can be selected. Template can be rotated to right 0, 90, 180, 270 degree. The rotation angle should not considered by summing. Rotation process is performed by taking 0 point is a reference. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

6. VERTICAL RIBBONSAVE/COLUMN NO: This feature provides to more than one print in same row. Printer prints up the selected number. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
7. VERTICAL RIBBONSAVE/SHIFTING: This feature provide to shift between two print out in same row after ribbonsave mode/column no is entered more than one. The unit is mm. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

There are 2 options to use the vertical ribbonsave function.

Option 1;
Template must be prepared based on the top left corner and direction of template must be 90°.

Option 2;
Template must be prepared based on the upper right corner and direction of template must be 0°.



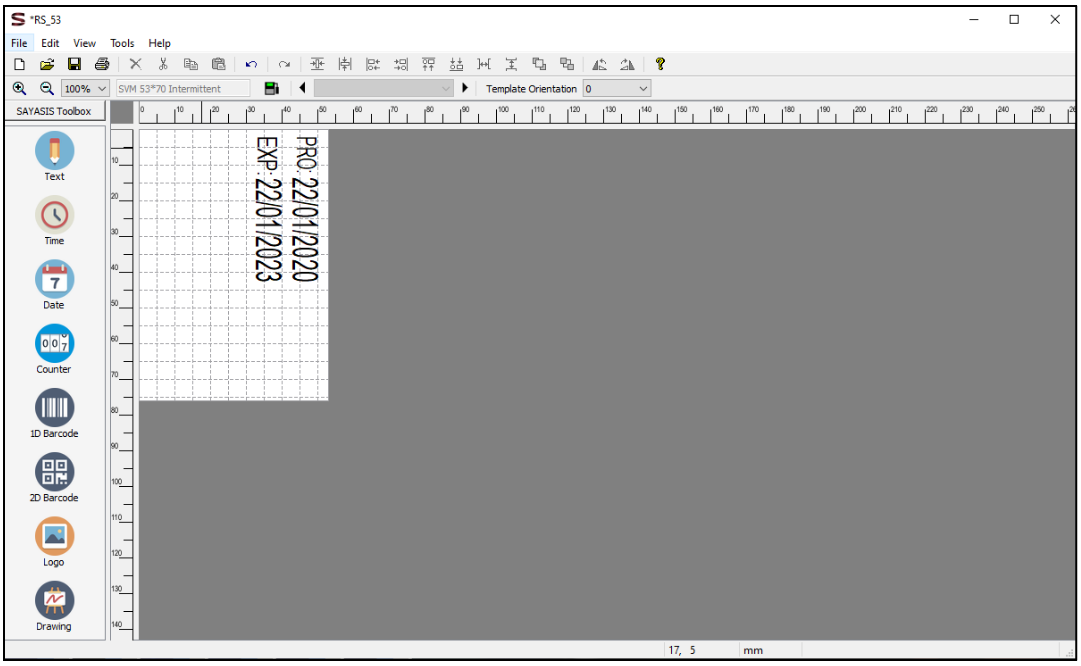


Figure 8. Preparation of Template Suitable for Vertical Ribbonsave Function - Option 1 (UP) and Option 2 (DOWN)

The following illustration (Figure 9) is drawn for VERTICAL RIBBONSAVE/COLUMN NO is 3. The distance between prints on ribbon is automatically adjusted according to the value entered in VERTICAL RIBBONSAVE/SHIFTING. If vertical ribbonsave function is used, the same column of ribbon is used for 3 prints since the VERTICAL RIBBONSAVE/COLUMN NO is entered 3. The point to note here is that if the function is used, printing on the package is not continuously done in the same place.

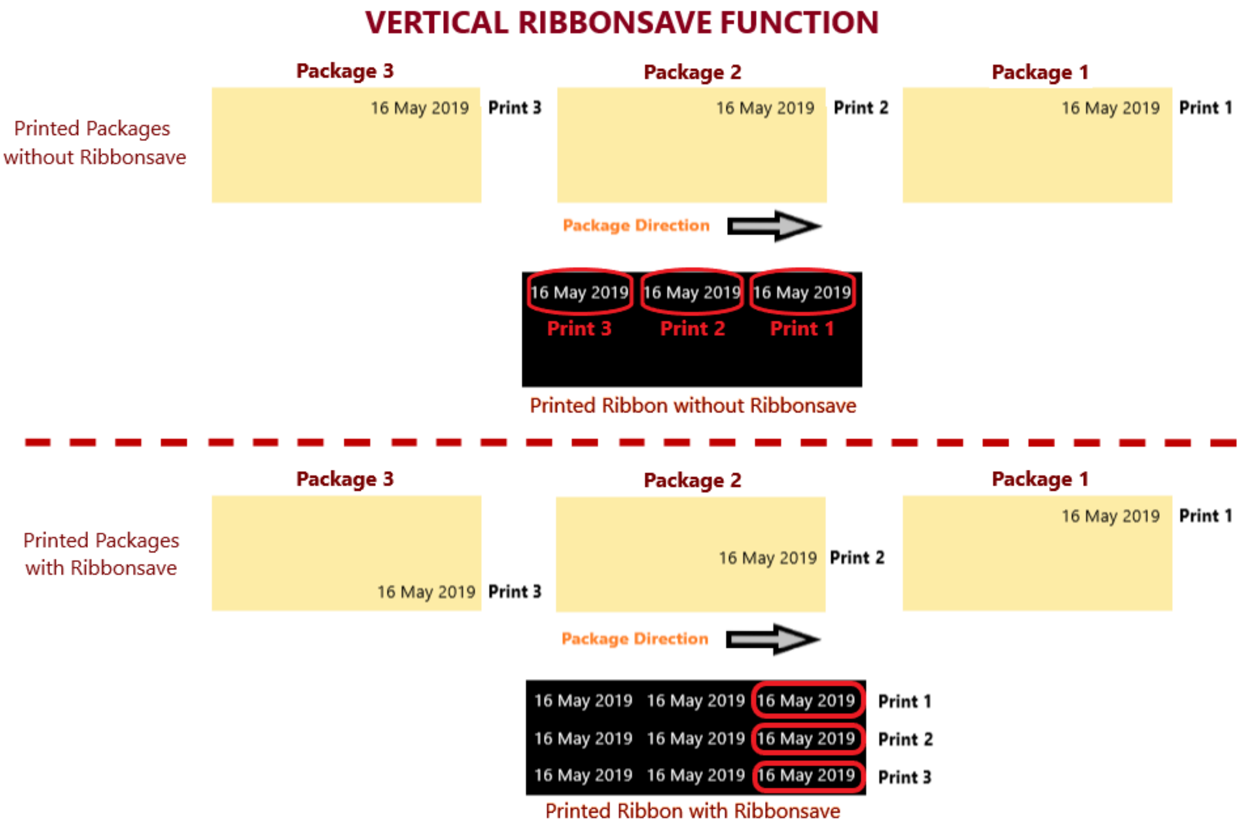


Figure 9. Vertical Ribbonsave Function

8. MIRRORING: This function provides to mirror template. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

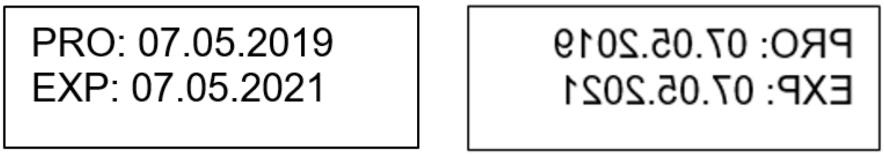


Figure 10. Original Template (Left) and Mirrored Template (Right)

9. HOME POSITION: This is used to adjust thermal print head print position. “**+**” button shifts to right. “**–**” button shifts to left. The unit is mm. The maximum shifting must be set according to the template. Otherwise it will cause ribbon breakage. For example; if 25 is entered, thermal print head prints after 25 mm. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.
If you add 10mm home position to the template it will move like the 2nd template

No	Designation
1	MAIN TEMPLATE
2	TEMPLATE WITH HOME POSTION ADDED

Table 5 Home Position

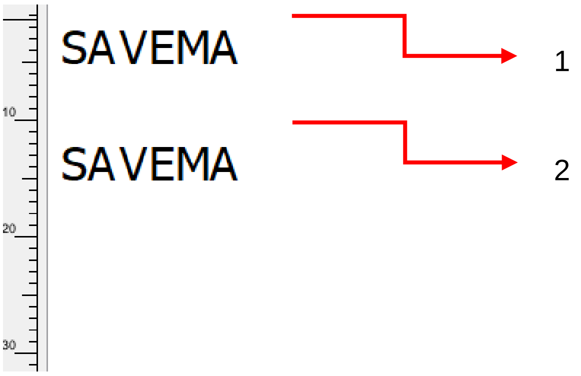


Figure 11 Home Position

10. HORIZONTAL RIBBONSAVE/COLUMN NO: This feature provides to more than one print in same row. Printer prints up the selected number. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

11. HORIZONTAL RIBBONSAVE/SHIFTING: This feature provide to shift between two print out in same row after ribbonsave mode/column no is entered more than one. The unit is mm. The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

The following illustration (Figure 12) is drawn for HORIZONTAL RIBBONSAVE/COLUMN NO is 2. The distance between prints on ribbon is automatically adjusted according to the value entered in HORIZONTAL RIBBONSAVE/SHIFTING. If horizontal ribbonsave function is used, the ribbon comes back for 1 print since the HORIZONTAL RIBBONSAVE/COLUMN NO is entered 2. The horizontal ribbonsave parameters must be precisely adjusted to the range in the print, otherwise a decrease in print quality may occur.

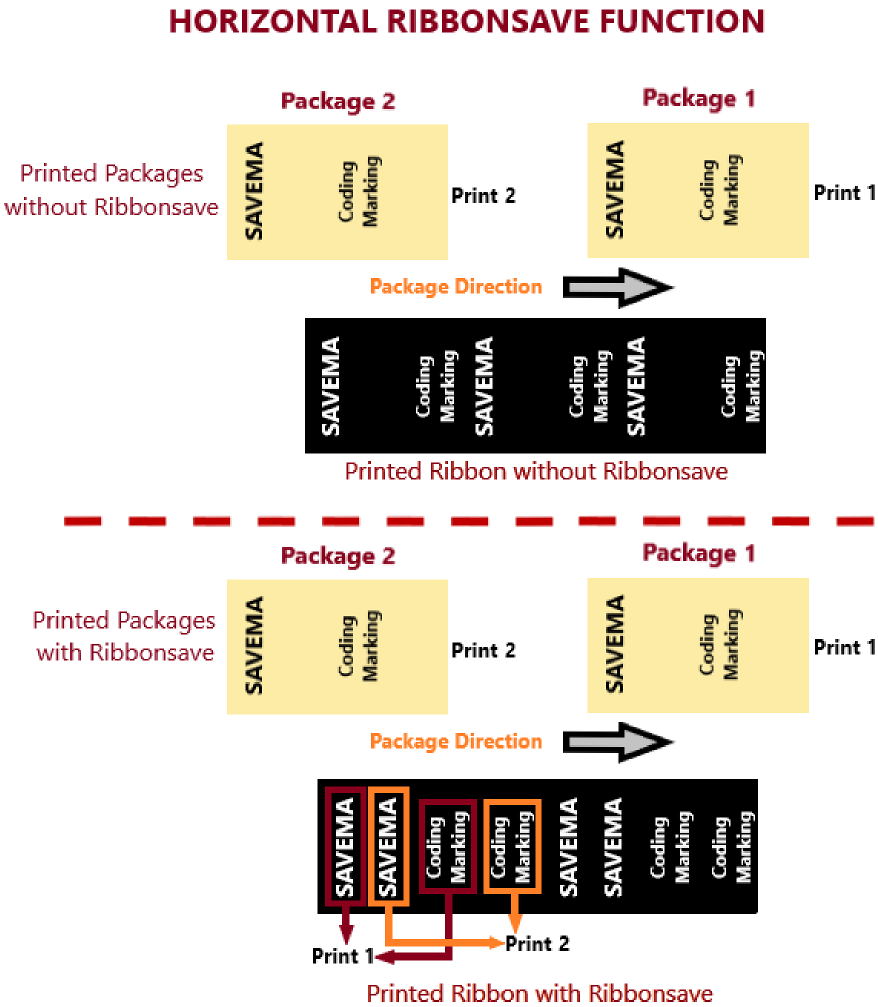


Figure 12. Horizontal Ribbonsave Function

12. OPERATING BUTTONS: The **APPLY** button must be pressed after the change has been made. Otherwise the change is not saved. The **EXIT** button must be pressed after the change has been saved. After the change is saved, the change will be applied in the first print.

1.1.4.2 CONTROL – CONTROL SECTION

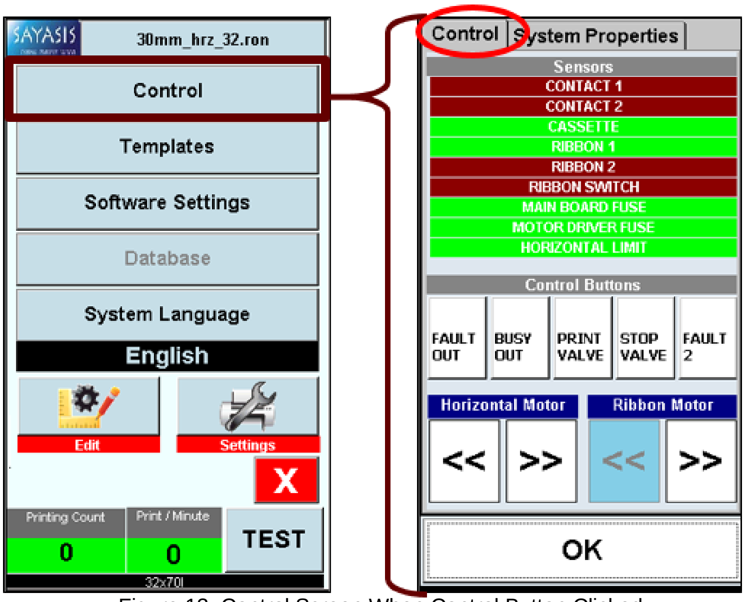


Figure 13. Control Screen When Control Button Clicked

Printer sensors, fuses, valves and motors can be controlled in this section. Also some signals that come to printer can be observed from the window. This function is not active while printer is printing. The printer must be stopped to enable this feature.

- CONTACT 1: Shows whether or not there is contact 1 signal. This field is for illustration only. System does not respond when clicked. If there is a contact signal, the field's color is green. Otherwise, color is red.
- CONTACT 2: Shows whether or not there is contact 2 signal. This field is for illustration only. System does not respond when clicked. If there is a contact signal, the field's color is green. Otherwise, color is red.
- CASETTE: Shows whether or not there is cassette. This field is for illustration only. System does not respond when clicked. If the cassette is inserted, the field's color is green. Otherwise, color is red.
- RIBBON 1: This is used to check if the ribbon is broken. This field is for illustration only. System does not respond when clicked. If the ribbon is broken. The field's color is red. Otherwise, color is green.
- RIBBON 2: Shows that the ribbon is turning left or right. This field is for illustration only. System does not respond when clicked. If the ribbon is turning, the field's color is green. Otherwise, color is red.
- RIBBON SWITCH: Shows whether the ribbon break switch is working. When the switch is pressed, the color turns green.
- MAINBOARD FUSE: Shows whether the main board fuse is damaged. This field is for illustration only. System does not respond when clicked. If the fuse is damaged, the field's color is red. Otherwise, color is green.
- MOTOR DRIVER FUSE: Shows whether the main board fuse is damaged. This field is for illustration only. System does not respond when clicked. If the fuse is damaged, the field's color is red. Otherwise, color is green.
- HORIZONTAL LIMIT SENSOR: Shows whether the horizontal limit sensor is working. When the sensor is detected, the color turns green.
- FAULT OUT: This used to check the printer fault relay. The button is darker when clicked and relay connections that are normally open becomes normally close state.
- BUSY OUT: This used to check the printer busy relay. The button is darker when clicked and relay connections that are normally open becomes normally close state.

- PRINT VALVE: This used to test whether the print piston is working. The button is darker when clicked and thermal print head goes to print position. **ATTENTION**: **Finger should not be placed under the thermal head while this feature is used.**

- STOP VALVE: This used to test whether the stop piston is working. The button is darker when clicked and thermal print head goes as far as it goes. **ATTENTION**: **Fingers should not be placed under the thermal head while this feature is used.**

- FAULT 2: This used to check the printer fault mosfet. The button is darker when clicked and the circuit is completed through mosfet. (See 2.2.3 for more detail about fault mosfet connection)

- RIBBON MOTOR: This used to detect if the ribbon motor is turning left or right. The button is darker when right direction button was clicked and ribbon motor turns right. Also the button is darker when left direction button was clicked and ribbon motor turns left.

- HORIZONTAL MOTOR: This used to detect if the horizontal motor is moving left or right. The button is darker when right direction button was clicked and motor moves thermal print head to right. Also the button is darker when left direction button was clicked and motor moves thermal print head to left.

1.1.4.3 CONTROL – INFORMATION SECTION

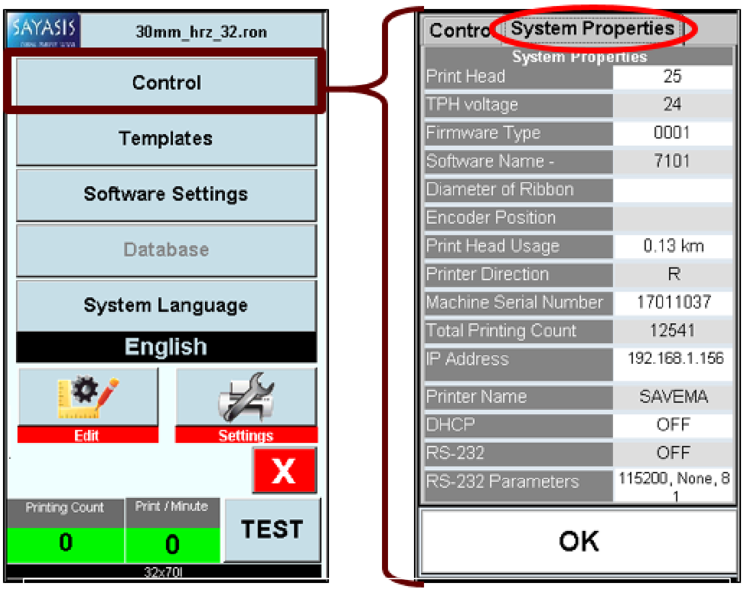


Figure 14. Information Screen When Control Button Clicked

This field provides to take some information about machine. These fields are for illustration only. System does not respond when clicked.

- PRINT HEAD TEMPERATURE (°C): This field gives information about momentary print head temperature.

- TPH VOLTAGE: This field gives information about momentary print head voltage.

- FIRMWARE TYPE: This field gives information about machine firmware.

- SOFTWARE NAME: The name of the software installed on the system.

- DIAMETER OF RIBBON: This field gives information about used ribbon diameter.

- PRINT HEAD USAGE: This field gives information about amount of thermal head used in km from the first print. The value is never reset.

- PRINTER DIRECTION: This field gives information about printer direction (**L**eft/**R**ight).

- MACHINE SERIAL NUMBER: This field gives information about machine serial number.

- TOTAL PRINTING COUNT: Shows the number of prints from the first print. The value is never reset, neither the machine is reset nor the template is changed.

- IP ADDRESS: This field gives information about machine IP address.

- PRINTER NAME: This field gives information about name of machine.
- DHCP: This field gives information about automatic IP address acquisition (**ON**/**OFF**).
- RS-232: This field gives information about connection of RS-232 (**ON**/**OFF**).
- RS-232 PARAMETER: This field gives information about RS-232 parameters.

OK BUTTON: Provides to return to the menu function screen.

1.1.5 HIDDEN SETTINGS

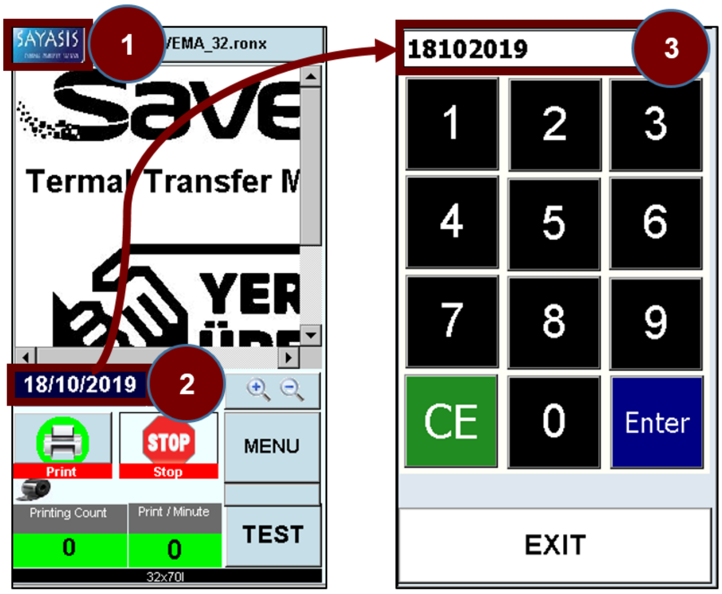


Figure 15. Password Screen When SAYASIS Button Clicked

No	Designation
1	SAYASIS LOGO
2	SYSTEM DATE
3	PASSWORD AREA

Table 6. Hidden Settings Signs

When the clicked on SAYASIS logo at main menu **[1]**, system wants to enter password that is system date. Password format is “dmyyyy”. *Example 1; password is “18102019” for this date “18/10/2019”. Example 2; password is “952019” for this date “09/05/2019”.* After password is entered correctly (after entering the value in **[2]** into **[3]**), the hidden settings screen is opened. The screen provides to adjust parameter that variables to the general work of the machine.

1.1.5.1 PARAMETRIC SETTINGS

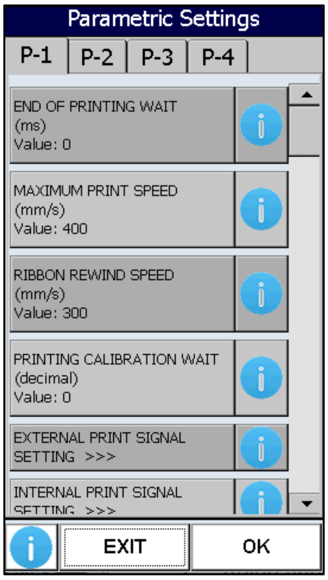


Figure 16. Parameter Menu

There are variables to the general work of the machine in this section. These settings provide the opportunity to make improvements in printing with relevant parameters.

VALUE CHANGE SYSTEM:

1. Select parameter by scrolling through pages and change the value by numeric numbers.
2. Push the “**APPLY**” button and **“OK”** button at right bottom corner sequentially.
3. Push **“OK”** button and **“EXIT”** button at right bottom corner sequentially.

Note: If “**EXIT”** button is clicked before changes haven’t been saved, there is no change in parameters.

Page NO	Parameter Name	Sub-parameter Name	Minimum	Default	Maximum	Unit
1	1) Horizontal Motor Lock Setting		0	**0**	1	decimal
	2) End of Printing Wait		0	**0**	900	ms
	3) Maximum Print Speed		50	**400**	500	mm/s
	4) Ribbon Rewind Speed		150	**300**	400	mm/s
	5) Printing Calibration Wait		0	**0**	150	decimal
	External Print Signal Settings	6) External Print Signal Source	0	**0**	1	decimal
		7) External Print Signal Min. Duty Time	0	**2**	100	ms
	Internal Print Signal Settings	8) Internal Print Signal Activation	0	**0**	1	decimal
		9) Internal Print Signal Package Length	1	**10**	500	ms*200
	Multi-Print Signal Settings	10) Multi-Print Signal Activation	0	**0**	1	decimal
		11) Multi-Print Signal Period	2	**2**	100	decimal
		12) Multi-Print Signal Package Length	1	**10**	500	ms*200
	Ribbon Selection	17) Ribbon Selection	0	**1**	1	decimal
		Ribbon Outer Diameter	34	**76**	99	mm
		Ribbon Inner Diameter	25	**33**	35	mm
		Ribbon Length	300	**600**	1000	m
2	18) TPH Resistance Value		1050	**1200**	1475	ohm
	19) Print Modes		0	**0**	8	decimal
	20) Ribbon Space Setting		50	**200**	500	mm/10
	21) Printer Type		0	**0**	7	decimal
	22) Printer Cassette Type		0	**0**	1	decimal
	23) TPH Stroke (Pressing) Time		0	**27**	500	ms
	Ribbon Hanging Roller Settings	24) Ribbon Rubber Roller Diameter 1	20	**32**	60	mm
		25) Ribbon Rubber Roller Diameter 2	0	**0**	99	mm/100
	26)TPH Voltage Calibration		50	**128**	150	decimal
	Pulley Roller Settings	27) Pulley Roller Diameter 1	10	**26**	50	mm
		28) Pulley Roller Diameter 2	0	**0**	99	mm/100
3	Ribbon Sensor Settings	29) Ribbon Broken Sensor Activation	0	**0**	3	decimal
		30) Ribbon Broken Sensor Limit	1	**4**	100	pulse
	Ribbon Tension Sensor Settings	31) Ribbon Tension Sensor Activation	0	**0**	1	decimal
		32) Ribbon Tension Sensor Limit	1	**4**	100	pulse
	Pre-Heating Settings	33) Pre-Heating Activation	0	**0**	1	decimal
		34) Pre-Heating Set Value	24	**24**	40	celsius
	35) Fuse Warning Setting		0	**0**	7	decimal
	36) Outputs Setting		0	**0**	7	decimal
4	Print Faulties Settings	37) Print Signal Fault Activation	0	**0**	2	decimal
		38) Print Signal Fault Limit Length	1	**10**	1000	ms*200
	39) Update Fault Activation		0	**0**	2	decimal
	42) TPH Temperature Calibration		1	**4**	100	decimal

Table 7. Parameters

PARAMETER SETTINGS PAGE-1

HORIZONTAL MOTOR LOCK SETTING:

When the cassette is removed from the printer, the horizontal motor moves itself to a certain position. If it is not desired to move after reaching this position, the motor can be locked with this parameter. If “0” is chosen, horizontal motor is unlocked, If “1” is chosen horizontal motor is locked. Default value is “0”.

END OF PRINTING WAIT:

When printing process is done, printer waits as this value before rewind home position. The waiting time cannot be less than 0 ms and more than 900 ms. Standard value is 0 ms for this parameter.

MAXIMUM PRINT SPEED:

This parameter determines the maximum speed that the machine will reach. The speed cannot be more than 500 mm/s and less than 50 mm/s. Standard speed value is 400 mm/s for this parameter.

RIBBON REWIND SPEED:

This parameter determines the maximum speed that the thermal head will reach when returning. The speed cannot be more than 400 mm/s and less than 150 mm/s. Standard speed value is 300 mm/s for this parameter.

PRINTING CALIBRATION WAIT:

The value cannot be more than 150 and less than 0. Standard value is 0 for this parameter.

EXTERNAL PRINT SIGNAL SETTINGS:

A. External Print Signal Source:

This parameter is used to choose contact type. If “0” is chosen, contact 1 is active contact 2 is passive. If “1” is chosen contact 1 is passive, contact 2 is active. Default value is “0”.

NOTE: External signal, internal signal and multi-print signal cannot be used at the same time. Only one of them can be used.

If contact 1 is selected, +24 V DC must be supplied to the contact 1 terminal on the contact wire. The GND of 24 V DC must also be connected to contact 1 GND on the contact wire. The same procedure is similar for contact 2. If contact 2 is selected, +24 V DC is connected to contact 2, and GND of 24 V is connected to contact 2 GND.

B. External Print Signal Min. Duty Time:

Contact signal duty time is not smaller than this value to print. Otherwise, no printing process. Standard duty time is 2 ms. The duty time cannot be more than 100 ms and less than 0 ms. This parameter is used to prevent undesired signal. Following sample is valid for the value is equal to 2 ms.

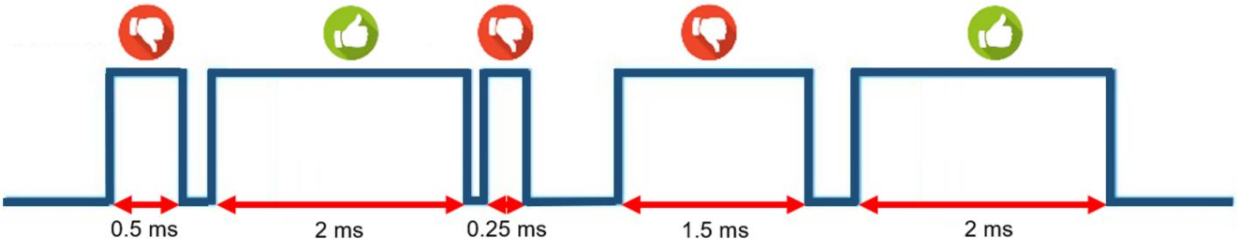


Figure 17. External Print Signal Minimum Duty Time Sample

INTERNAL PRINT SIGNAL SETTINGS:

A. Internal Print Signal Activation:
This parameter is used to activate internal print function. If “0” is chosen, the function is passive. If “1” is chosen, the function is active. Default value is “0”.
NOTE: External signal, internal signal and multi-print signal cannot be used at the same time. Only one of them can be used.

B. Internal Print Signal Package Length:
Printer takes contact signal from internal signal source depend on pack length if the parameter is activated. Internal print signal package length indicates time in ms between sequential two prints. The value entered here must be greater than the time it takes to print a template. Internal print signal minimum package length cannot be lower than 1 [ms*200] and upper than 500 [ms*200]. Standard length is 10 [ms*200]. Following sample is valid for the value is equal to 200.

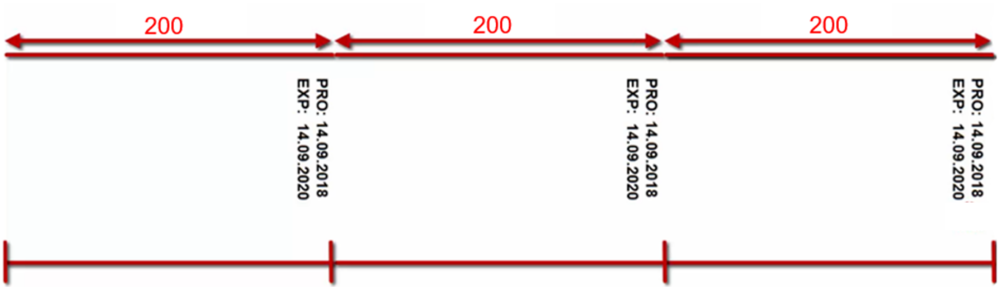


Figure 18. Internal Print Sample

MULTI-PRINT SIGNAL SETTINGS:

A. Multi-Print Signal Activation:
This parameter is used to activate multi print function. If “0” is chosen, the function is passive. If “1” is chosen, the function is active. Default value is “0”.

NOTE: External signal, internal signal and multi-print signal cannot be used at the same time. Only one of them can be used.

B. Multi-Print Signal Period:
Multi-Print signal period is number of print in an external print signal. The period cannot be more 100 and less than 2. Standard is 2 for this parameter.

C. Multi-Print Signal Package Length:
Printer takes contact signal from trigger signal source depend on package length if the parameter is activated. Multi-Print signal package length is indicates time in ms between sequential two prints. The package length parameter cannot be lower than 1 [ms*200] and upper than 500 [ms*200]. Standard length is 10 [ms*200].

The following sample is valid for multi-print signal period is equal to 3 and multi-print signal package length is 200. So 5*200 = 1000 ms and it means one second.

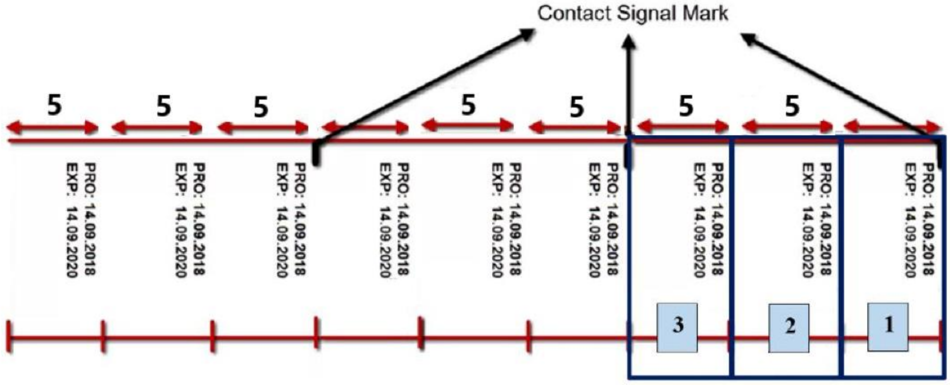


Figure 19. Multi-Print Sample

RIBBON SELECTION;

A. Ribbon Selection: There are two modes. Mode 0 is for high speed ribbon and mode 1 is for normal speed ribbon. The mode can be changed according to the type of ribbon used. Mode 0 is selected if the ribbon used supports high speed. Default mode is 1.

B. Ribbon Outer Diameter: Ribbon outer diameter cannot be more than 99 mm and less than 34 mm. Standard diameter is 76 mm.

C. Ribbon Inner Diameter: Ribbon inner diameter cannot be more than 35 mm and less than 25 mm. Standard diameter is 33 mm. When measuring the inner diameter of the ribbon, neglect the roller it is wound on.

D. Ribbon Length: Ribbon length cannot be more than 1000 m and less than 300 m. Standard diameter is 600 m. Ribbon length cannot be selected more than 600 m if printer is not XL model.

PARAMETER SETTINGS PAGE-2

TPH RESISTANCE VALUE:

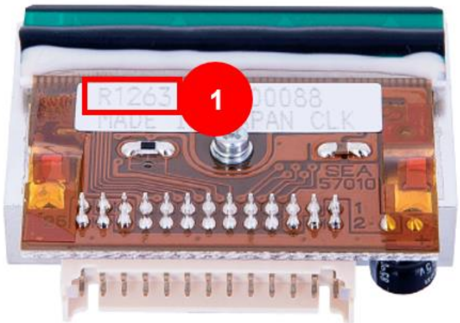


Figure 20. Thermal Print Head

Thermal print head resistance value(R) written on TPH label. The resistance cannot be lower than 1050 Ω and upper than 1475 Ω.

No	Designation
1	TPH RESISTANCE VALUE

Table 8. Thermal Print Head Description

PRINT MODES:

There are 9 print modes. These modes change according to dot for a pixel, maximum speed and ribbon material.

Mode	Pixel	Ribbon Type	Maximum Speed
0	ONE DOT FOR A PIXEL	WAX-RESIN RIBBON	500
1	ONE DOT FOR A PIXEL	RESIN RIBBON	500
2	Not used	Not used	Not used
3	DOUBLE DOT FOR A PIXEL	WAX-RESIN RIBBON	500
4	DOUBLE DOT FOR A PIXEL	RESIN RIBBON	500
5	Not used	Not used	Not used
6	TRIPLE DOT FOR A PIXEL	WAX-RESIN RIBBON	500
7	TRIPLE DOT FOR A PIXEL	RESIN RIBBON	500
8	Not used	Not used	Not used

Table 9. Print Modes

The mode should be chosen according to the required criteria. Since the printer automatically adjusts to the selected mode, the mode must be selected according to the ribbon type used. Otherwise, print quality may be poor. Mode 0-3-6 are used for wax-resin ribbon and have same print quality. Mode 1-4-7 are used for resin ribbon and have same print quality. Standard mode is “0”.

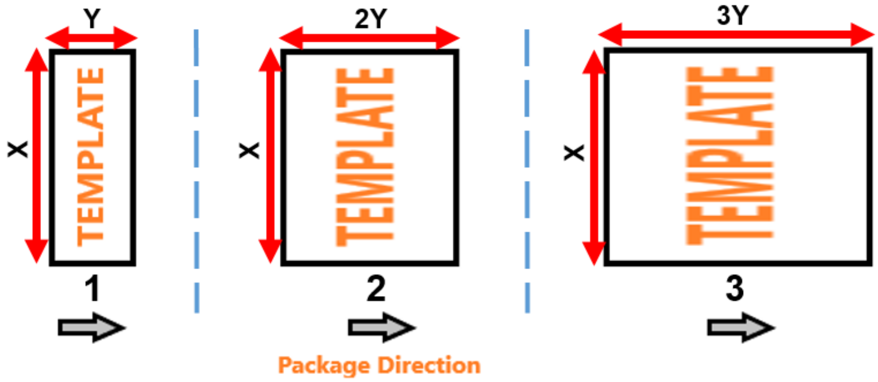


Figure 21. Dot for a Pixel Samples

No	Designation (Template Dimensions)
1	ONE DOT FOR A PIXEL
2	DOUBLE DOT FOR A PIXEL
3	TRIPLE DOT FOR A PIXEL

Table 10. Dot for a Pixel Types

RIBBON SPACE SETTING:

This value decreases the gap of two print-out on ribbon. Each 10 motor pulses affect 1mm. If the value is less than 200, ribbon space decreases. If the value is more than 200, ribbon space increases. The space value cannot be less than 50 [mm/10] and more than 500 [mm/10]. Standard is 200 [mm/10] for this parameter.

PRINTER TYPE:

Mode	Printer Type	Mode	Printer Type
0	32 mm	4	32 mm XL
1	53 mm	5	53 mm XL
2	107 mm	6	107 mm XL
3	128 mm	7	128 mm XL

Table 11. TPH Type Modes

This parameter is used to declare to system the type of printer used. There are 8 printer type and default value is entered according to buying printer type.

PRINTER CASSETTE TYPE:

This parameter is used to declare to system whether the device is cassette. If “0” is entered, that’s mean the device has cassette. If “1” is entered, that’s mean the device has no cassette. Standard type is entered according to buying machine cassette type.

TPH STROKE (PRESSING) TIME:

This parameter is used to adjust TPH reach time from piston position to rubber roller position after contact signal was received. The time cannot be more than 500 ms and less than 0 ms. Standard is 27 ms for this parameter.

RIBBON HANGING ROLLER SETTING:

A. Ribbon Rubber Roller Diameter 1:

Diameter of rubber roller should be entered. The diameter cannot be more than 60 mm and less than 20 mm. Standard is 40 mm for this parameter.

B. Ribbon Rubber Roller Diameter 2:

Percentage value of rubber roller diameter should be entered. The diameter cannot be more than 99 mm/100 and less than 0 mm/100. Standard is 0 mm/100 for this parameter.

For example; if ribbon feeding rubber roller diameter is 30.50 mm, ribbon feeding rubber roller diameter 1 should be entered 30 and ribbon feeding rubber roller diameter 2 should be entered 50.

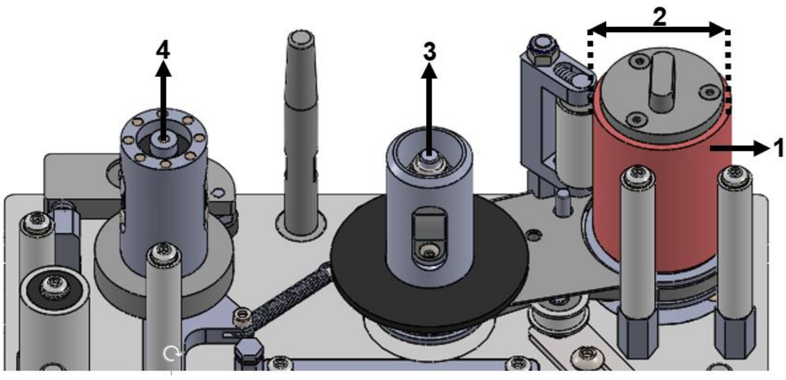


Figure 22. Rubber Roller Diameter Representation

No	Designation
1	RIBBON RUBBER ROLLER
2	RIBBON RUBBER ROLLER DIAMETER
3	USED RIBBON
4	UNUSED RIBBON

Table 12. Description of Rubber Roller Representation

TPH VOLTAGE CALIBRATION:

Mainboard factory setting should not change without technical personal. TPH resistance value must be “1200” and ribbon thickness setting must be “1” for calibration. TPH voltage can adjust as 24V with this value. TP3 and TPGND test points on the mainboard are used to measure voltage. The measurement is made by placing the “+” end of the multimeter to the TP3 and the “-” end to the TPGND. The calibration value cannot be less than 50 and more than 150. Standard is 128 for this parameter.

PULLEY ROLLER SETTING:

A. Pulley Roller Diameter 1:

Diameter of pulley roller should be entered. The diameter cannot be more than 50 mm and less than 10 mm. Standard is 26 mm for this parameter.

B. Pulley Roller Diameter 2:

Percentage value of pulley roller diameter should be entered. The diameter cannot be more than 99 mm/100 and less than 0 mm/100. Standard is 0 mm/100 for this parameter.

For example; if pulley roller diameter is 20.75 mm, pulley roller diameter 1 should be entered 20 and pulley roller diameter 2 should be entered 75.

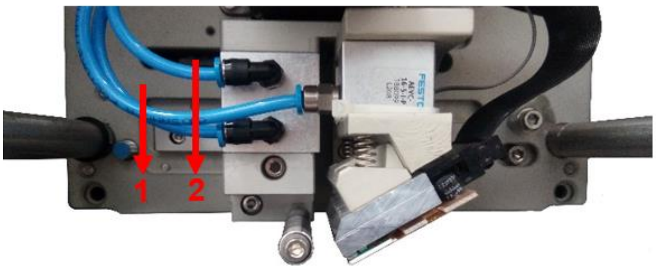


Figure 23. Pulley Roller and Motion Belt Representation

No	Designation
1	PULLEY ROLLER
2	RIGHT LEFT MOTION BELT

Table 13. Description of Pulley Roller and Motion Belt Representation

PARAMETER SETTINGS PAGE-3

RIBBON SENSOR SETTING:

A. Ribbon Broken Sensor Activation: There are 4 modes and default value is “0”;

Mode		
0	RIBBON BREAK ACTIVE	RIBBON NOT FOUND ACTIVE WITH MAGNETS
1	RIBBON BREAK PASSIVE	RIBBON NOT FOUND ACTIVE WITH MAGNETS
2	RIBBON BREAK PASSIVE	RIBBON NOT FOUND PASSIVE WITH MAGNETS
3	RIBBON BREAK ACTIVE	RIBBON NOT FOUND ACTIVE WITH SWITCH

Table 14. Ribbon Broken Sensor Activation Modes

B. Ribbon Broken Sensor Limit:
When mode 0 is selected, ribbon broken alarm occur after sensor limit value. The limit cannot be more than 100 pulse and less than 1 pulse. Standard limit is 4 pulse for this parameter.

RIBBON TANSION SENSOR SETTING:

A. Ribbon Tension Sensor Activation:
This parameter is used to activate ribbon tension sensor. If “0” is chosen, the sensor is active with magnets. If “1” is chosen, the sensor is passive. Default value is “0”.

B. Ribbon Tension Sensor Limit:
When sensor is activated, ribbon tension alarm occur after limit of magnets value. The limit cannot be more than 100 pulse and less than 1 pulse. Standard limit is 4 pulse for this parameter.

PRE-HEATING SETTING:

A. Pre-Heating Activation:
This parameter is used to activate pre-heating. If “0” is chosen, the pre-heating is active. If “1” is chosen, pre-heating is passive. Default value is “0”.

B. Pre-Heating Set Value:
Pre-heating system controls temperature. This value determine minimum TPH temperature. The value cannot be more than 40 °C and less than 24 °C. Standard value is 24 °C.

FUSE WARNING SETTING:
Default value is “0” and there are 8 modes (MB: Main Board - MD: Motor Driver);

Mode			
0	MB FUSE WARNING ACTIVE	MD FUSE WARNING ACTIVE	TPH ERROR ACTIVE
1	MB FUSE WARNING PASSIVE	MD FUSE WARNING ACTIVE	TPH ERROR ACTIVE
2	MB FUSE WARNING ACTIVE	MD FUSE WARNING PASSIVE	TPH ERROR ACTIVE
3	MB FUSE WARNING PASSIVE	MD FUSE WARNING PASSIVE	TPH ERROR ACTIVE
4	MB FUSE WARNING ACTIVE	MD FUSE WARNING ACTIVE	TPH ERROR PASSIVE
5	MB FUSE WARNING PASSIVE	MD FUSE WARNING ACTIVE	TPH ERROR PASSIVE
6	MB FUSE WARNING ACTIVE	MD FUSE WARNING PASSIVE	TPH ERROR PASSIVE
7	MB FUSE WARNING PASSIVE	MD FUSE WARNING PASSIVE	TPH ERROR PASSIVE

Table 15. Fuse Modes

If MB fuse warning is activated, printer gives error when mainboard fuse is damaged. In this case the fuse on mainboard should be controlled and changed if necessary. In changing situation, pay attention to the value of the fuse. Detailed information on the location and value of the mainboard fuse is discussed in section 4.1.1.

If MD fuse warning is activated, printer gives error when motor driver fuse is damaged. In this case the fuse on motor driver should be controlled and changed if necessary. In changing situation, pay attention to the value of the fuse. Detailed information on the location and value of the motor driver fuse is discussed in section 4.1.2.

If TPH error is activated, there are 3 situation that cause to error.

1. When TPH temperature is less than 1 °C and more than 55 °C.
2. When TPH is not installed.

3. When TPH cable is broken or not fully inserted.

OUTPUTS SETTING:
Default value is “0” and there are 8 fault output modes;

Mode		
0	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL PASSIVE(FC)
1	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL PASSIVE(FC)
2	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL ACTIVE(FC)
3	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL ACTIVE(FC)
4	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL PASSIVE(HC)
5	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL PASSIVE(HC)
6	MACHINE GIVES ALARMS AT ERROR TIMES AND PASSIVE TIME	BUSY SIGNAL ACTIVE(HC)
7	MACHINE GIVES ALARM AT ERROR TIMES.	BUSY SIGNAL ACTIVE(HC)

Table 16. Alarm Modes

- FC means full cycle. Printer gives busy signal while TPH is in print process and during come back to initial state.
- HC means half cycle. Printer gives busy signal while TPH is in print process but doesn’t give busy signal during come back to initial state.
- ERROR times mean an error has occurred in the system like ribbon break etc.
- Passive time means the printer is at stop position not print position.

PARAMETER SETTINGS PAGE-4

PRINT FAULTIES SETTING:

A. Print Signal Faulties Activations:
This parameter is used to adjust print signal error is active or passive. When adjusting the print signal error setting, warning and fault out settings are also made. Warning means if there is an error, the error appears on screen. Fault out means determines whether the system gives error output even if there is fault warning. Default value is “0” and there are 3 modes;

Mode	Warning	Fault Out	Print Signal Error
0	PASSIVE	PASSIVE	PASSIVE
1	ACTIVE	ACTIVE	ACTIVE
2	ACTIVE	PASSIVE	ACTIVE

Table 17. Print Signal Fault Modes

Warning: If warning property is activated, warning related activated print signal error occur on screen. When warning is not activated, even if error occurs user does not see error warning on screen.

Fault out: If fault out property is activated, printer gives error output related activated print signal error. When fault out is not activated, even if error occurs printer does not give output.

Print Signal Error: If print signal error property is activated, printer gives printer signal error when contact signal does not received during the time entered in print signal faulty limit length parameter.

B. Print Signal Faulty Limit Length:
When print signal error is activated, printer wait print signal to not give error as print signal fault limit. If print signal is not received in print signal faulty limit length, printer gives print signal error. The

length cannot be more than 1000 [ms*200] and less than 1 [ms*200]. Standard length is 10 for this parameter.

UPDATE FAULT ACTIVATION:

This parameter is used to adjust update error is active or passive. When adjusting the update error setting, warning and fault out settings are also made. Warning means if there is an error, the error appears on screen. Fault out means determines whether the system gives error output even if there is fault warning. Default value is “0” and there are 3 modes;

Mode	Warning	Fault Out	Update Error
0	PASSIVE	PASSIVE	PASSIVE
1	ACTIVE	ACTIVE	ACTIVE
2	ACTIVE	PASSIVE	ACTIVE

Table 18. Updatable Area Fault Modes

Warning: If warning property is activated, warning related activated update error occur on screen. When warning is not activated, even if error occurs user does not see error warning on screen.

Fault out: If fault out property is activated, printer gives error output related activated update error. When fault out is not activated, even if error occurs printer does not give output.
Update Error: If update error property is activated, printer gives update error when variable template doesn’t change. Variable template is like number of package, serial number that specific to each product etc.

TPH TEMPERATURE CALIBRATION:

This parameter is used to read print head temperature calibration value. The calibration value cannot be less than 1 and more than 100. Standard is 4 for this parameter.

1.2 ERRORS

1.2.1 COMMUNICATION ERRORS

After cable installation; turn power on/off switch to “ON” mode. When you turn the power switch, firstly open screen will come and if all the connections well, screen goes to next window. If there is a communication problem, screen shows message on the left. This error is not dependent on to the parameter, so it cannot be set to be passive or active with parameter.

When the error occurs, check steps in below;

- The communication cable between the screen and the printer must be fully inserted into the sockets and the cable must be unbroken.
- The communication cable between mainboard and screen board must be fully inserted into the sockets and the cable must be unbroken.
- The screen board must be unbroken (there should be no burn or discharge marks around any component).
- The little board in the screen must be fully inserted into the socket.

If the communication between the display and the printer suddenly stops after the printer has started, the following warning appears

Rs422 communication error

When the error occurs, check steps in below;

- The communication cable between the screen and the printer must be fully inserted into the sockets and the cable must be unbroken.
- The communication cable between mainboard and screen board must be fully inserted into the sockets and the cable must be unbroken.

1.2.2 RIBBON BREAK/RIBBON NOT FOUND ERROR

The machine has a switch that detects the ribbon break and sensors that detect the absence of ribbon in the system. These elements inform the printer in the event of any problem. If there is a ribbon break problem, screen shows message on the bottom.

Existing Ribbon broken. Please change the Ribbon.

When ribbon break error occurs, check steps in below;

- Air pressure should not exceed 2–2.5 bars (maximum).
- Ribbon mounting must be correct. If not, there can be a ribbon tension problem. It must be checked according to the instructions on the cartridge cover.
- Check the ribbon tension. If it is not good, then determine the reason and correct the problem.
- The ribbon may be wrapped around the rubber roll. Investigate to see if there is any problem with ribbon conveyance.
- If the thermal print head printing edge is not parallel to the printing surface, then the ribbon may break.
- The ribbon broken sensor cable on the mainboard must be fully seated and not broken.

If there is a ribbon not found problem, screen shows message on the bottom.

Ribbon not found. Please insert ribbon.

When ribbon not found error occurs, check steps in below;

- Ribbon should be replaced according to the instructions on the cartridge cover.
- The ribbon broken sensor cable and switch cable on the mainboard must be fully seated and not broken.

Activation of ribbon break and ribbon not found errors can be adjust in the parameter menu. Ribbon sensor setting/Ribbon broken sensor activation parameter dedicated to this work.

1.2.3 TPH ERROR

Thermal print head (TPH) is the part of the printer used to print. Tph-related problems may occur in some cases. In this case the warning message on the bottom will appear.

Thermal print head error. Please check below steps. 1- Print head cable must be inserted well. 2- Print head temperature must be between +1 and +55 °C. 3- Print head must be robust. 4- Print head cable must be robust. If problem continue, please contact with manufacturer.

Activation of TPH error can be adjust in the parameter menu. Fuse warning setting parameter dedicated to this work.

When TPH error occurs, check steps in below;

- Tph must be installed in the system.
- Tph cable should not be broken
- Tph cable must be fully inserted into the TPH cable connection board socket.
- The cable must be fully inserted into the print head cable socket.
- Optimal temperature must be 24 °C. Pre-heating setting/pre-heating activation parameter should be “0” for automatically heating.

1.2.4 MAINBOARD/MOTOR DRIVER FUSE ERROR

Fuses are used in electronic circuit boards to protect devices after a certain current. If there is a mainboard fuse problem, screen shows message on the bottom.

Main board fuse is broken. Please contact manufacturer.

If there is a motor driver problem, screen shows message on the bottom.

Motor driver fuse is broken. Please contact manufacturer.

When fuse error occurs, check steps in below;

- Make sure the fuses are not damaged. The location of the mainboard fuse is shown in section 4.1.1 and the location of the motor driver board fuse is shown in section 4.1.2. If there is damage, it must be replaced.

Activation of fuse errors can be adjust in the parameter menu. Fuse warning setting parameter dedicated to this work. The fuses can be checked for damage in the control menu (for detail see section 1.1.4.2).

1.2.5 UPDATE ERROR

Some templates change with each print. These are called variable templates. Variable template is like number of package, serial number that specific to each product etc. Printer gives update error when the printer prints the same print twice. If there is an update problem, screen shows message on the bottom.

Update error

When update error occurs, check steps in below;

- Be sure to use a variable template. If the template does not have this property, update error property should be turned off.
- Communication may be slow if there are too many variables in the same template. In this case the print speed should be reduced.

Activation of update error can be adjust in the parameter menu. Update fault activation parameter dedicated to this work.

1.2.6 PRINT SIGNAL ERROR

The printer needs a contact signal to print. Printer gives print signal error if the printer does not receive contact signal within the value entered in print faults setting/print signal faulty limit length parameter after switching the printer to print mode. If there is a print signal problem, screen shows message on the bottom.

Print signal error

When print signal error occurs, check steps in below;

- The value entered in the print faults setting/print signal faulty limit length parameter must be greater than the packet length used.
- The contact cable must be fully inserted into the sockets and the cable must be unbroken.
- Check the contact signal to the printer.
- The contact board must be unbroken (there should be no burn or discharge marks around any component).

Activation of print signal error can be adjust in the parameter menu. Print faults setting/Print faults activations parameter dedicated to this work.

1.2.7 TEMPLATE ERROR

When the machine is first turned on, the printer does not have a template under normal state. The screen shows following warning to install a template on the printer.

Please install the template.

When template error occurs, check steps in below;

- Make sure template is installed.

1.2.8 MEMORY ERROR

If there is a problem ram on the mainboard, screen shows warning on the bottom.

Memory error

When memory error occurs, check steps in below;

- Ram may be corrupted. Please contact manufacturer.

1.2.9 MOTOR DRIVER COMMUNICATION ERROR

When there is a communication problem between motor driver and mainboard, screen shows following warning.

Motor driver communication error

When motor driver communication error occurs, check steps in below;

- Make sure communication cable between motor driver and mainboard is fully inserted.

1.2.10 VOLTAGE ADJUSTMENT ERROR

When TPH voltage is lower than 24 °C, screen shows warning on the bottom.

Voltage adjustment error

When voltage adjustment error occurs, check steps in below;

- TPH resistance value is entered “1200” on parameter menu.
- Ribbon thickness setting is entered “1”.
- TP3 and TPGND test points on the mainboard are measured. The measurement is made by placing the “+” end of the multimeter to the TP3 and the “-” end to the TPGND.
- TPH voltage calibration value is changed until the value on the meter indicates 24 volts.
- TPH resistance value is rechanged to original TPH resistance.

1.2.11 RIBBON TENSION ERROR

The ribbon must has a certain tension for the printer to print. When the ribbon has not optimal tension, screen shows warning on the bottom.

Ribbon tension error

When ribbon tension error occurs, check steps in below;

- Air pressure should not exceed 2–2.5 bars (maximum).
- Prevent any vibration caused by the packaging machine.
- Make sure the pistons are working correctly.
- Make sure the head is in the correct position (more detail see section 3.5.1).
- Make sure TPH temperature is 24 °C.

1.2.12 COVER ERROR

When the printer cover is not fully inserted, the following message is shown on the screen.

Printer Cover is Open. Please Close the Cover.

When cover error occurs, check steps in below;

- Gently pull the cover outward and push it straight into the printer.

1.2.13 HORIZONTAL LIMIT SENSOR ERROR

The following error appears when communication between the printer and sensor cannot be established.

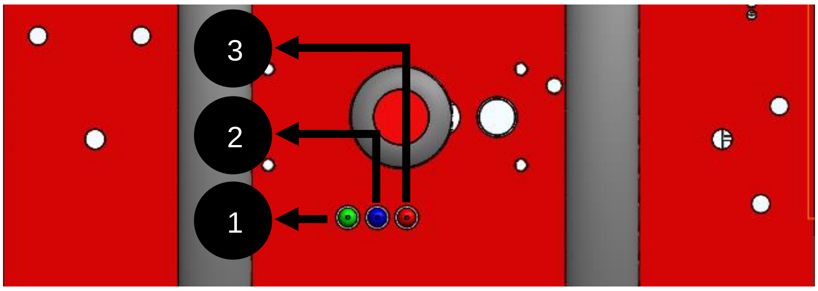
Horizontal limit sensor is broken. Please contact manufacturer.

When horizontal limit sensor error occurs, check steps in below;

- Make sure that the limit sensor is fully inserted into the socket on the motor driver.
- Make sure communication cable between motor driver and mainboard is fully inserted.
- If the problem is not resolved in the previous steps, contact manufacturer.

1.3 CASSETTE LED DESCRIPTIONS

There are LEDs on the cassette which provide information about various situations. The color of the LEDs may vary from product to product. The reference point is the location of the printer. On a flat-installed printer, the leftmost LED indicates communication, the middle LED indicates fault, the rightmost LED indicates contact.



No	Designation
1	COMMUNICATION LED
2	FAULT LED
3	CONTACT LED

Table 19. LEDs Descriptions on Cassette

Figure 24. Information LEDs on Cassette

1. Communication LED: This LED indicates communication between the screen and the printer when the machine is first turned on. If the LED is blinking during opening, there is no problem in communication. There is a led used for the same operation on the mainboard. In order to control that led, it is necessary to remove the case of the machine. The LED on the cassette is designed to eliminate this difficulty.

This LED is used for another operation after the machine has completed the opening process. Another operation is that the LED is on as long as the thermal head is in the printing position.

2. Fault LED: This LED is on when the printer gives an error. For proper operation of the LED, it should be ensured that the desired errors are activated in the parameters section.

3. Contact LED: This LED lights up when the printer receives a contact signal. The LED is designed to check whether the contact signal comes correctly. This led only works when contact 1 input is used. If the contact is given from the contact 2 input, the led will not operate.

2

ELECTRICAL DIAGRAMS

2.1 MACHINE ELECTRONIC COMPONENTS AND BOARDS CONNECTION

Basically, there are 4 boards in the printer;

- Mainboard
- Motor driver board
- Screen communication board (power board)
- Contact board

Besides, there is a connection board for TPH cable. Also there are 2 boards in screen;

- Screen board
- COM board

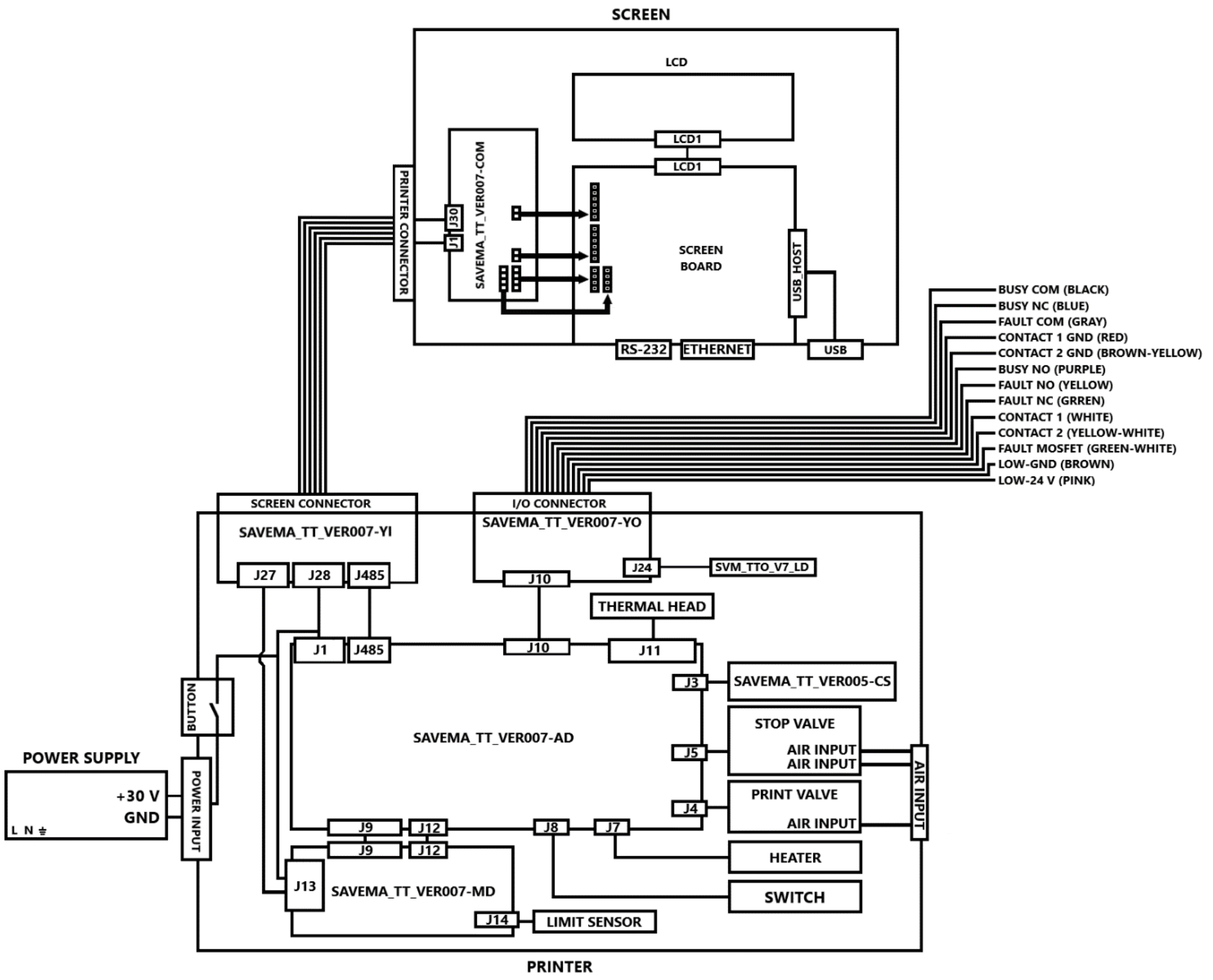


Figure 25. General Electrical Connections

2.2 PACK MACHINE CONNECTIONS

Contact signal is required for the printer to print. If this contact signal is to be supplied from an external source (such as a photocell or relay), attention must be paid to the type of source, ie whether it is PNP or NPN. Once the type of source has been determined, the connection can be made as shown below.

In addition, the printer generates fault and busy signals to be used in the package machine. The connections of these signals to the packaging machine are shown below.

There are 2 contact inputs for the printer. These two inputs cannot be used at the same time. Contact 2 is designed to be an alternative when the contact 1 input fails. Which contact to use must be activated in the parameters section. External print signal settings / external print signal source parameter dedicated to this work. Also, the printer cannot detect the signal from an inactive contact. In addition, if contact 1 is used, contact 1 GND must be used as GND. If contact 2 is used, contact 2 GND must be used as GND.

2.2.1 PNP CONNECTIONS

Photocell sensors have two different types, PNP or NPN. The connection diagram varies depending on the type of sensor. This is a point to be considered for the smooth operation of the system. The PNP type sensors trigger the positive voltage. Connection must be made according to this rule.

The relays have three ports; COM, NO, NC. When the relay is triggered by means of terminals A1+ and A2-, the signal given from terminal COM can be used from terminal NO. When the A1+ and A2- terminals are not triggered, the signal from the COM terminal is transmitted to the NC terminal. For example, if 24 V is connected to the COM terminal and the relay is triggered, there will be 24 V at the NO end and no signal at the NC. Similarly, if 24 V is connected to the COM terminal and the relay is not triggered, there will be 24 V at the NC terminal, and no signal will be at the terminal NO.

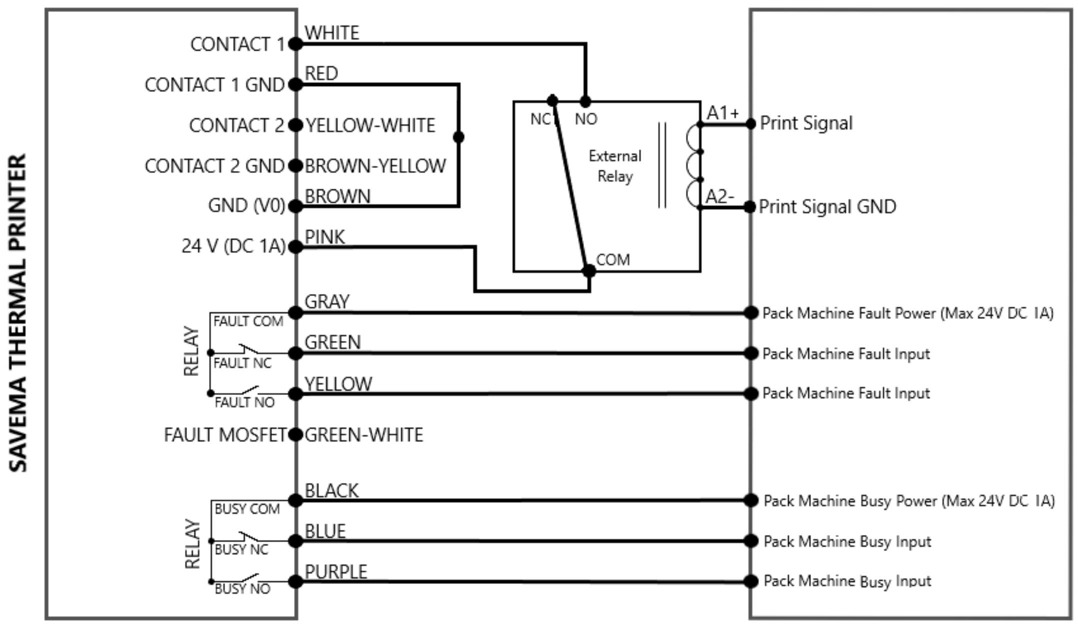


Figure 26. PNP Relay Connections

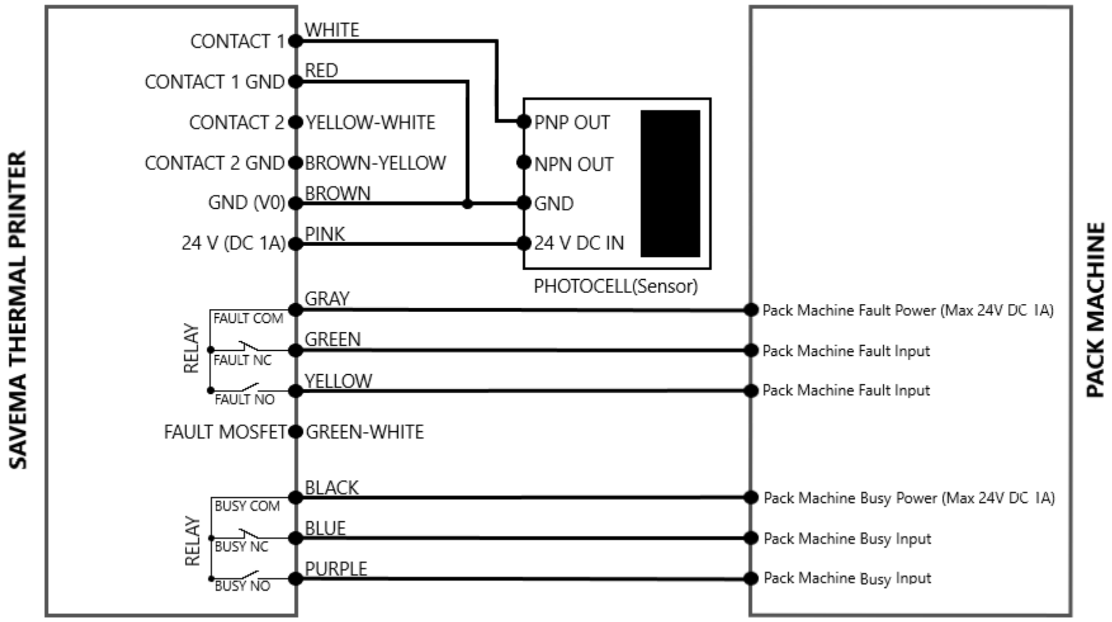


Figure 27. PNP Photocell Connections

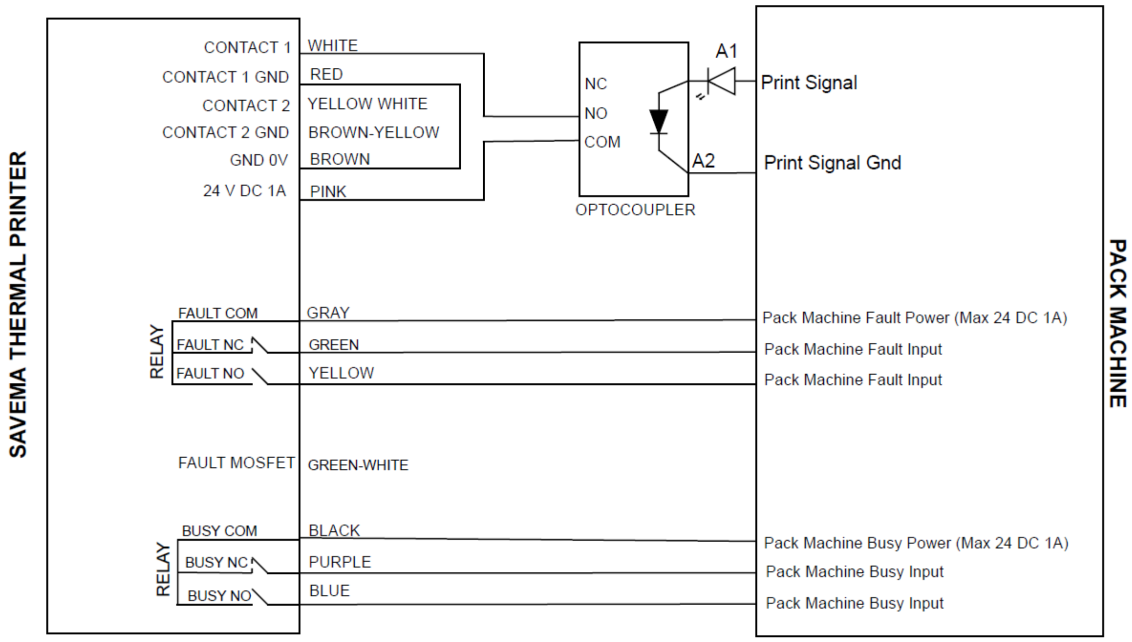


Figure 28 PNP Optocoupler Connections

2.2.2 NPN CONNECTIONS

Photocell sensors have two different types, PNP or NPN. The connection diagram varies depending on the type of sensor. This is a point to be considered for the smooth operation of the system. The NPN type sensors trigger the negative/neutral voltage. Connection must be made according to this rule.

The relays have three ports; COM, NO, NC. When the relay is triggered by means of terminals A1+ and A2-, the signal given from terminal COM can be used from terminal NO. When the A1+ and A2- terminals are not triggered, the signal from the COM terminal is transmitted to the NC terminal. For example, if GND is connected to the COM terminal and the relay is triggered, there will be 0 V at the

NO end and no signal at the NC. Similarly, if GND is connected to the COM terminal and the relay is not triggered, there will be 0 V at the NC terminal, and no signal will be at the terminal NO.

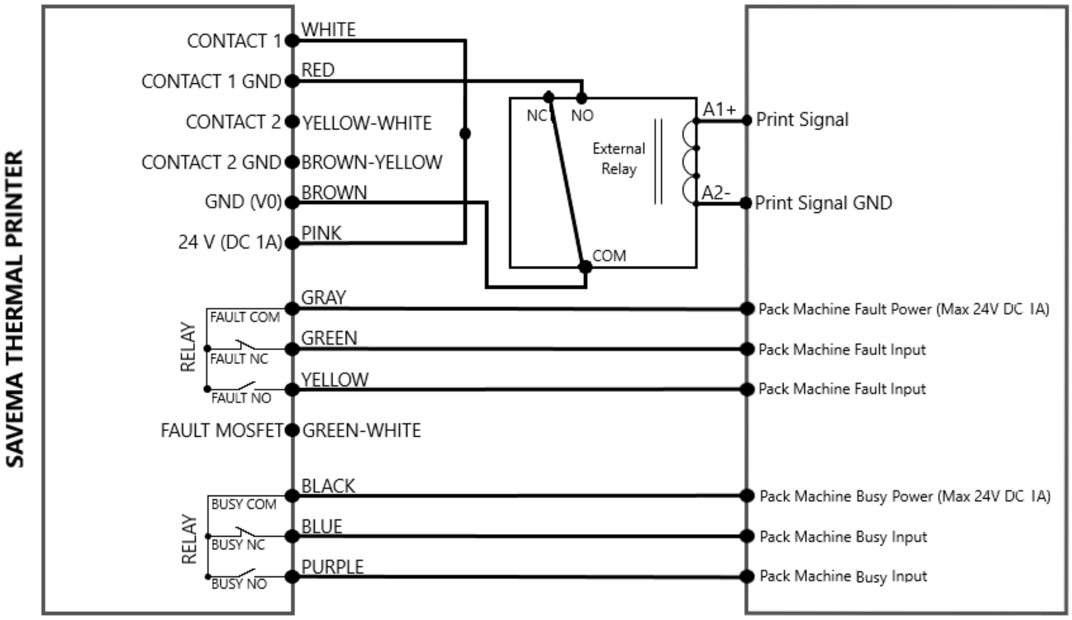


Figure 29. NPN Relay Connections

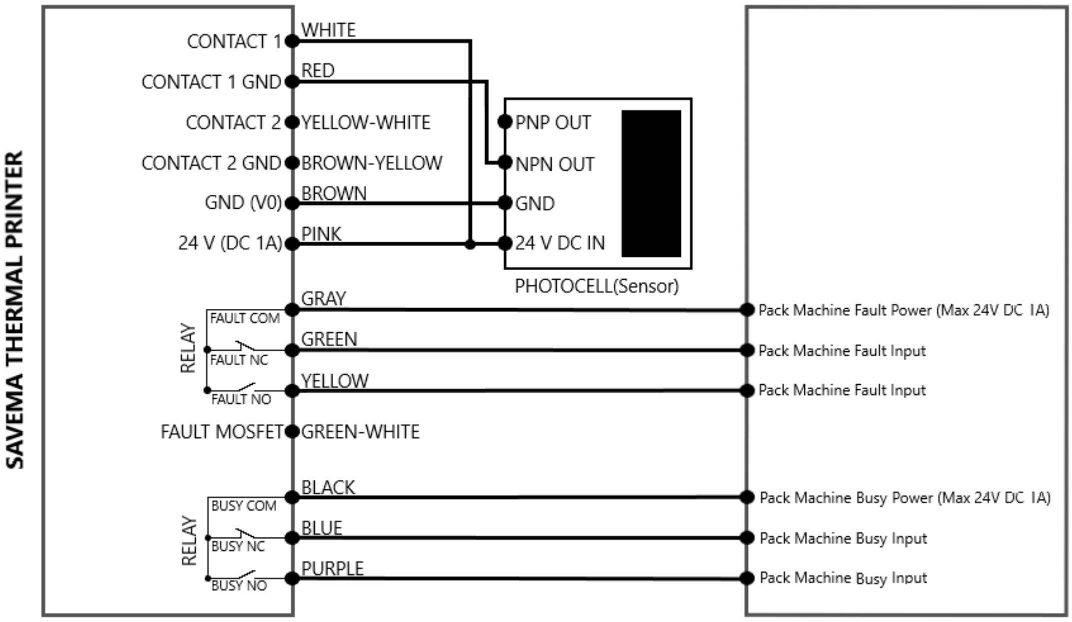


Figure 30. NPN Photocell Connections

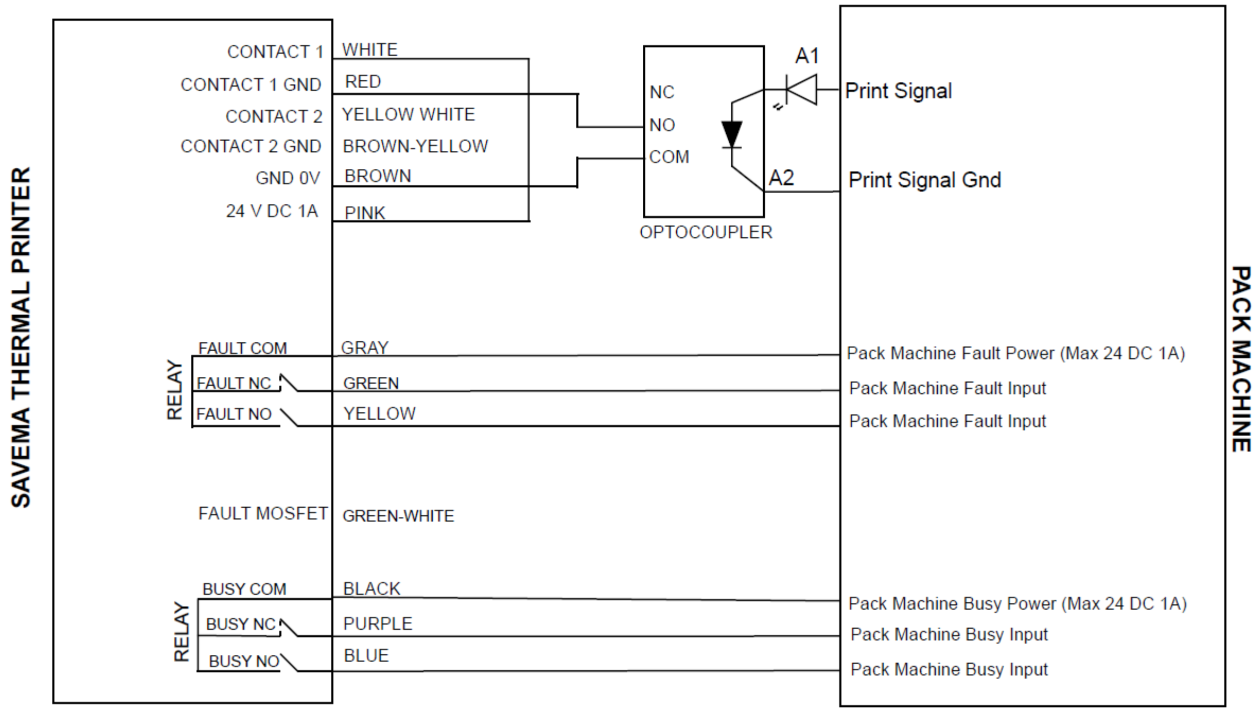


Figure 31 NPN Optocoupler Connections

2.2.3 FAULT MOSFET CONNECTION

Fault mosfet output can be used if fault signal output is wanted to receive the error signal from the mosfet rather than from the relay. 24 V on the printer's contact cable (pink color cable) to the A1+ pin of the external relay. The fault mosfet output on the printer's contact cable (green-white color cable) is also connected to A2- pin of the external relay.

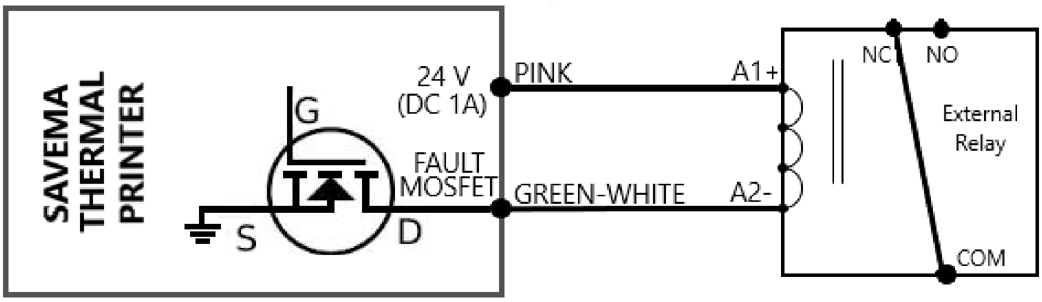


Figure 32. Fault Mosfet Connection

3 INSTALLATION

3.1 POWER CABLE

Power supply AC power input’s voltage can change according to country voltage value (110V or 220V). Make sure that the power supply you have purchased meets country standards.

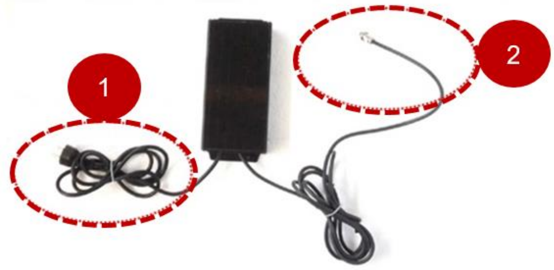


Figure 33. Power Supply
Spare Part Number: GK-32

No	Designation
1	POWER SUPPLY AC INPUT CABLE
2	POWER SUPPLY DC OUTPUT CABLE

Table 20. Power Supply Input and Output

DC voltage pins from power supply are shown below.

Figure 34. Power Supply DC Output Cable Terminal Numbers

No	Designation
1/2	+30 V DC
3	GND
4/5	-V (0 V)

Table 21. Power Supply Output Cable Pin-Outs

The DC output cable of the power supply is connected by a secondary cable that the other end of which is DB15 female connector.

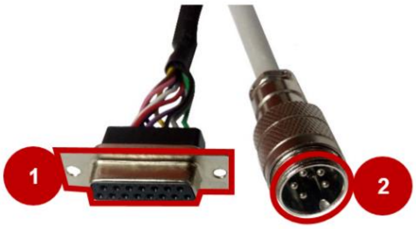


Figure 35. Power Supply Secondary Cable
Spare Part Number: GK-DB15

No	Designation
1	POWER SUPPLY SECONDARY CABLE TO PRINTER
2	POWER SUPPLY SECONDARY CABLE TO POWER SUPPLY

Table 22. Power Supply Secondary Cable Descriptions

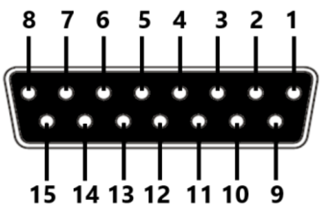


Figure 36. Power Supply Secondary Cable DB15 Terminal Numbers

No	Designation
1/2/10/11	30 V DC
4/5/13/14	-V (0 V)
7/15	GND

Table 23. Power Supply Secondary Cable DB 15 Pin-Outs

Figure 37. Power Supply Cable Connections

No	Designation
1	30 V DC
2	30 V DC
3	-V (0 V)
4	-V (0 V)
5	GND
6	NOTR
7	LINE

Table 24. Power Supply Cable Connections

3.2 PRINTER CABLES

The cables attached to the printer are gathered in a common socket. This common socket contains contact, screen and power supply sockets.

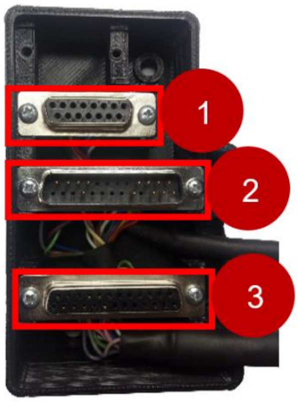


Figure 38. Printer Common Socket

No	Designation
1	FEMALE POWER SUPPLY SOCKET
2	MALE CONTACT SOCKET
3	FEMALE SCREEN SOCKET

Table 25. Printer Common Socket Descriptions

Figure 39. Screen Cable
Spare Part Number: C-DATAV7-43

No	Designation
1	SCREEN CABLE TO PRINTER
2	SCREEN CABLE TO SCREEN

Table 26. Screen Cable Descriptions

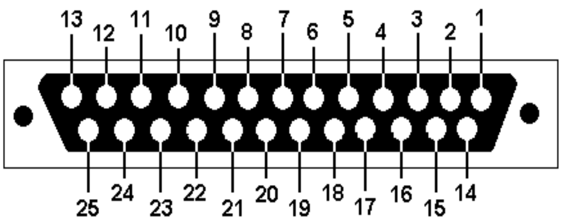


Figure 40. Screen Cable Printer Side Terminal Numbers

No	Designation
24	+30 V
16/17	GND
5	Rx +
18	Rx
6	Tx +
19	Tx
OTHERS	NO CONNECTION

Table 27. Screen Cable Printer Side Terminal Explanations

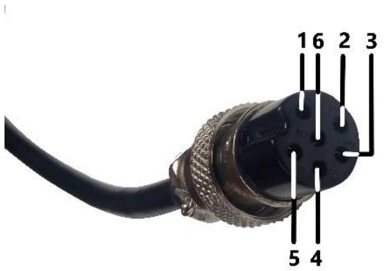


Figure 41. Screen Cable Screen Side Terminal Numbers

No	Designation
1	+30 V
2	GND
3	Rx +
4	Rx
5	Tx
6	Tx +

Table 28. Screen Cable Screen Side Terminal Explanations

One side of the contact cable is attached to the printer, while the other side is left to the user to connect it to the required places. The contact cable contains the cables carrying the busy, fault and contact signals.

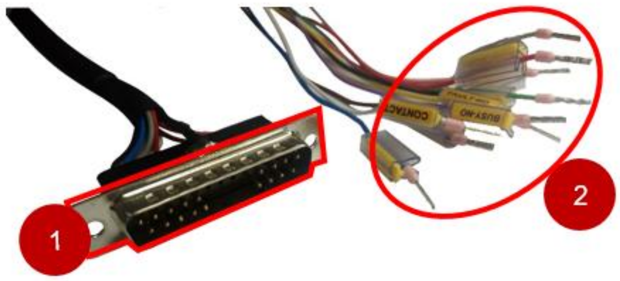


Figure 42. Contact Cable
Spare Part Number: C-CNTV7

No	Designation
1	CONTACT CABLE TO PRINTER
2	CONTACT CABLE TO USER

Table 29. Contact Cable Description

The contact cable that belong with “SVM20i Series 32*70 I” can be used on “SVM 32*70 I v6” printer but the contact 2, contact 2 GND and fault-mosfet outputs **must be kept closed** and **must not touch any place**. Otherwise, may cause serious damage to the printer.

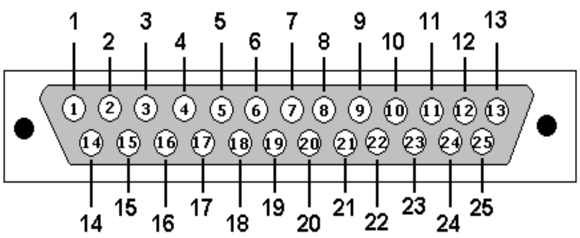


Figure 43. Contact Cable Printer Side Terminal Numbers

No	Designation	Explanation	Color(User Side)
1	BUSY COM	BUSY COM	BLACK
2	BUSY N.C.	BUSY RELAY NORMALLY-CLOSED CONTACT PIN	BLUE
3	FAULT COM	FAULT COM	GRAY
4	-	-	-
5	CONTACT 1 GND	GND 1 PIN OF THE PRINT SIGNAL TO WRITE	RED
6/7/8/9	-	-	-

10	CONTACT 2 GND	GND 2 PIN OF THE PRINT SIGNAL TO WRITE	BROWN-YELLOW
11/12/13	-		-
14	BUSY N.O.	BUSY RELAY NORMALLY-OPEN CONTACT PIN	PURPLE
15	FAULT N.O.	FAULT RELAY NORMALLY-OPEN CONTACT PIN	YELLOW
16	FAULT N.C.	FAULT RELAY NORMALLY-CLOSED CONTACT PIN	GREEN
17	CONTACT 1	CONTACT 1 PIN OF THE PRINT SIGNAL TO WRITE	WHITE
18/19/20/21	-	-	-
22	CONTACT 2	CONTACT 2 PIN OF THE PRINT SIGNAL TO WRITE	YELLOW-WHITE
23	FAULT MOSFET	FAULT MOSFET CONTACT PIN	GREEN-WHITE
24	LOW-GND	-	BROWN
25	LOW-24 V	24V DC (MAX 1A) OUTPUT	PINK

Table 30. Contact Cable Printer Side Terminal Explanations

BUSY COM: This pin must be connected to the busy power output of the packaging machine. Busy power output of the packaging machine must be maximum 24 volt DC and 1 ampere. This connection must be done to use the busy signal.
BUSY N.C.: This pin is active when the printer does not produce a busy signal. If it is active, it transmits the voltage that coming from the busy com to the output.
BUSY N.O.: This pin is active when the printer produces a busy signal. If it is active, it transmits the voltage that coming from the busy com to the output.
Busy Com, Busy N.C. and Busy N.O. connections are illustrated in section 2.

FAULT COM: This pin must be connected to the fault power output of the packaging machine. Fault power output of the packaging machine must be maximum 24 volt DC and 1 ampere. This connection must be done to use the fault signal.
FAULT N.C.: This pin is active when the printer does not produce a fault signal. If it is active, it transmits the voltage that coming from the fault com to the output.
FAULT N.O.: This pin is active when the printer produces a fault signal. If it is active, it transmits the voltage that coming from the fault com to the output.
FAULT MOSFET: This pin gives a fault signal output when printer produce a fault signal even if the fault relay fails.
Fault Com, Fault N.C., Fault N.O. and Fault Mosfet connections are illustrated in section 2.

CONTACT 1: The contact signal is required for the printer to print. If this signal is to be received from outside, the + pin of the signal must be connected to this pin.
CONTACT 1 GND: - pin of the signal that + pin is connected to contact 1 is connected to this pin.
Contact 1 must be selected from the parameters in order to use contact 1. External print signal setting/External print signal source parameter should be entered 0. (For more detail, see section 1.1.5.1). Contact 1 connections are mentioned in section 2.2.

CONTACT 2: Contact 2 is designed for use in the event of a contact 1 input failure. The + pin of the print signal must be connected to this pin. Contact 2 and contact 1 cannot be active at the same time. In order to use contact 2, External print signal setting/External print signal source parameter should be entered 1.
CONTACT 2 GND: - pin of the signal that + pin is connected to contact 2 is connected to this pin.
Contact 2 connections are same as contact 1 connections mentioned in section 2.2. That is, the connection to Contact 1 is made to Contact 2 and the connection to Contact 1 GND is made to Contact 2 GND.

LOW-24 V: It is the + pin of the supply power to the system to be used to generate the contact signal.
LOW-GND: It is the - pin of the supply power to the system to be used to generate the contact signal.
(See section 2.2 for Low-24 V and Low-GND connection diagram)

3.3 MACHINE I/O CONNECTION DIAGRAM

The cables attached to the machine's and screen's casing are shown below;

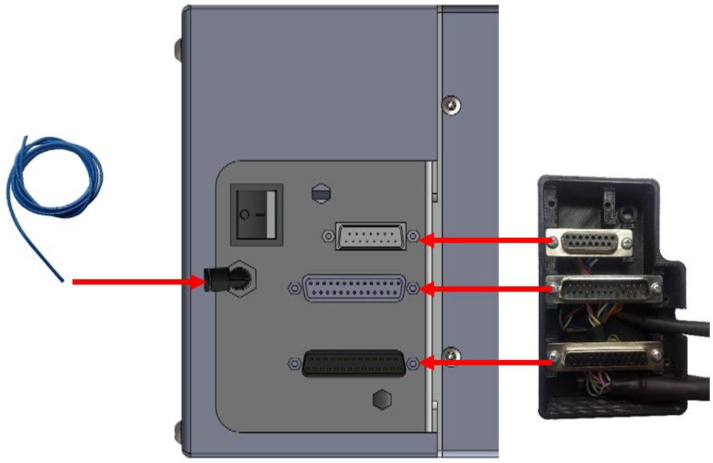


Figure 44. Contact, Screen, Power Cable Sockets from Printer Side

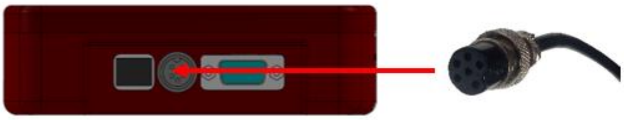


Figure 45. Screen Cable Socket from Screen Side

3.4 THERMAL PRINTER ELECTRONIC INSTALLATION

At the time of mechanical installation, make sure there is no power in the system when disconnecting cables. Remove all cables when it is clear that there is no power in the system.

3.4.1 TPH INSTALLATION

Distance of between Print Head and Print Surface must be 1.5mm and print head must be parallel to surface. Rear and top views of the thermal print head must be like bellow pictures;

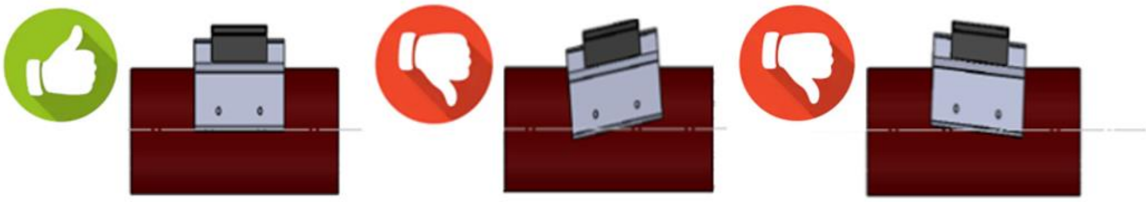


Figure 46. Top View of Thermal Print Head

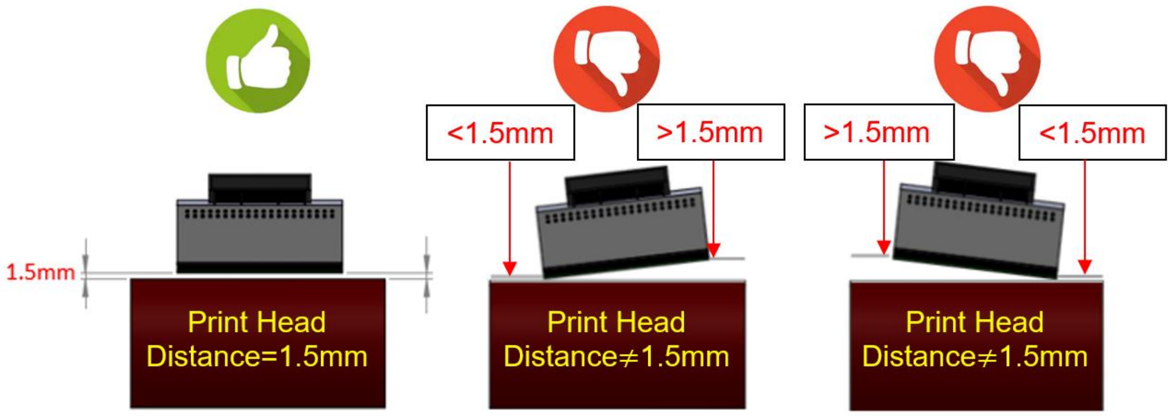


Figure 47. Rear View of the Thermal Print Head

After completing all installations, connect all cables and energize the system. Make sure enter the all cable well.

3.5 BOARD CHANGINGS

Screws that on the back of the printer must be removed to open for electronic part.

3.5.1 MAINBOARD CHANGING

Mainboard can be changed following the steps below.

1. Disconnect power connection.
2. Take off all cable in mainboard (shown in the picture with rectangle)
3. Remove the screws with Allen wrench (shown in the picture with round).
4. Insert a new board.
5. Replace the screws to removed place and insert all cables.

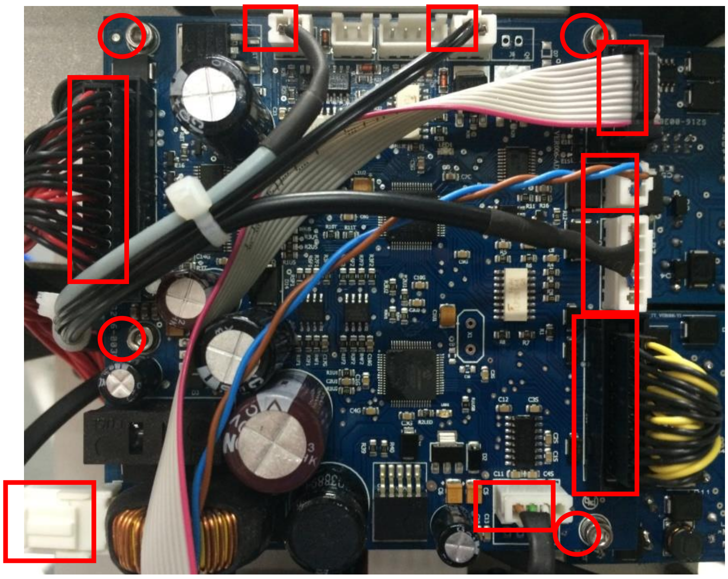


Figure 48. Mainboard Replacement Screws

3.5.2 MOTOR DRIVER BOARD CHANGING

Motor driver board can be changed following the steps below.

1. Disconnect power connection.
2. Take off all cable in motor driver board (shown in the picture with rectangle)
3. Remove the screws with Allen wrench (shown in the picture with round).
4. Insert a new board.
5. Replace the screws to removed place and insert all cables.

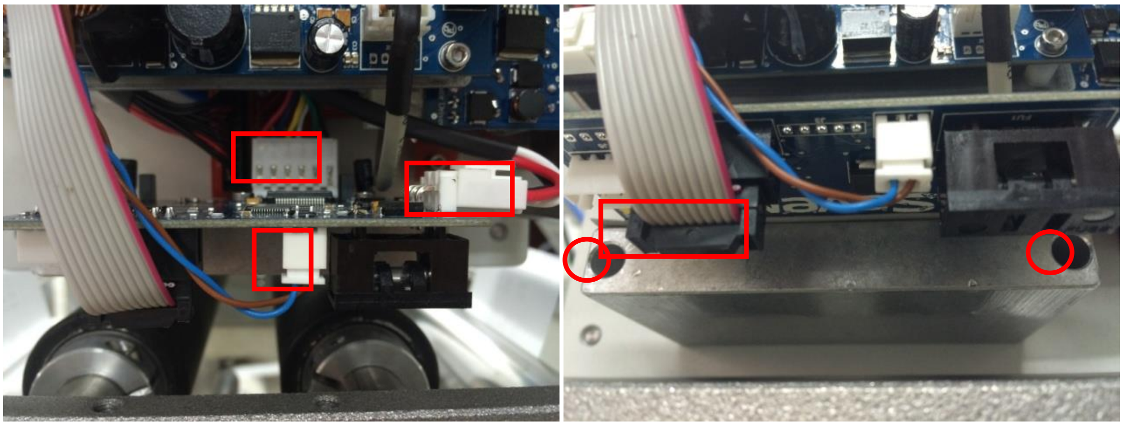


Figure 49. Motor Driver Board Replacement Screws

3.5.3 SCREEN COMMUNICATION BOARD CHANGING

Screen communication board can be changed following the steps below.

1. Disconnect power connection.
2. Take off screen connector cable from machine case (shown with dashed round).
3. Take off all cable in screen communication board (The board is shown in the picture with dashed rectangle and cables are shown with undashed rectangle).
4. Remove the screws with Allen wrench (shown in the picture with undashed round).
5. Insert a new board.
6. Replace the screws to removed place and insert all cables.

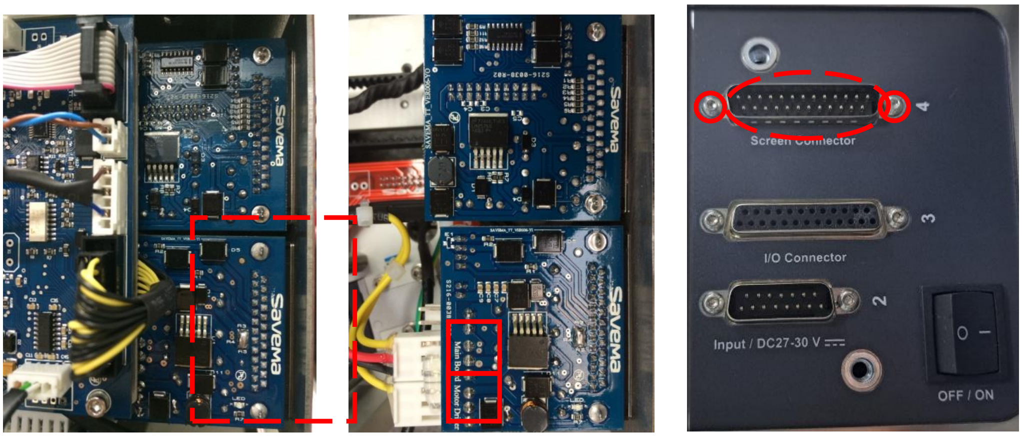


Figure 50. Screen Communication Board Replacement Screws

3.5.4 CONTACT BOARD CHANGING

Contact board can be changed following the steps below.

1. Disconnect power connection.
2. Take off I/O connector cable from machine case (shown with dashed round).
3. Take off all cable in contact board (The board is shown in the picture with dashed rectangle and cables are shown with undashed rectangle).
4. Remove the screws with Allen wrench (shown in the picture with undashed round).
5. Insert a new board.
6. Replace the screws to removed place and insert all cables.

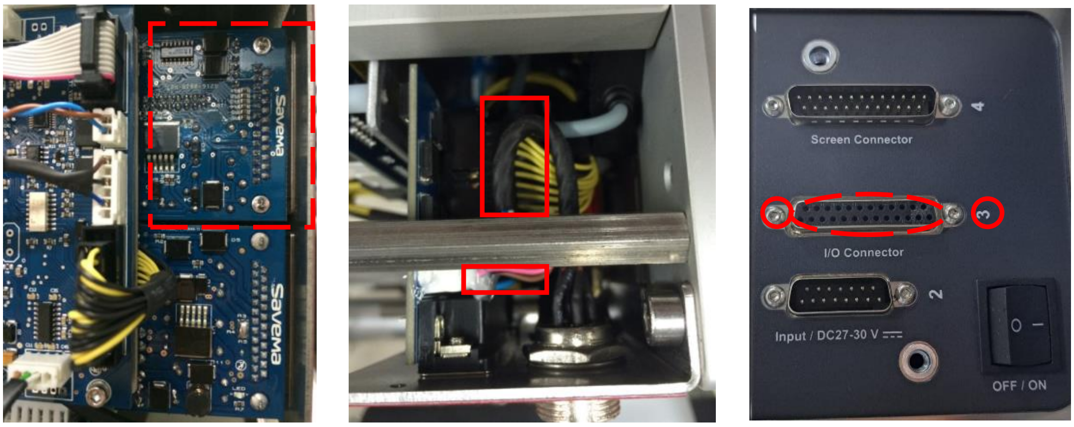


Figure 51. Contact Board Replacement Screws

4 TECHNICAL INFORMATION

4.1 BOARDS AND CONNECTIONS

The correct positions of the boards and the correct connections of the cables are shown below;

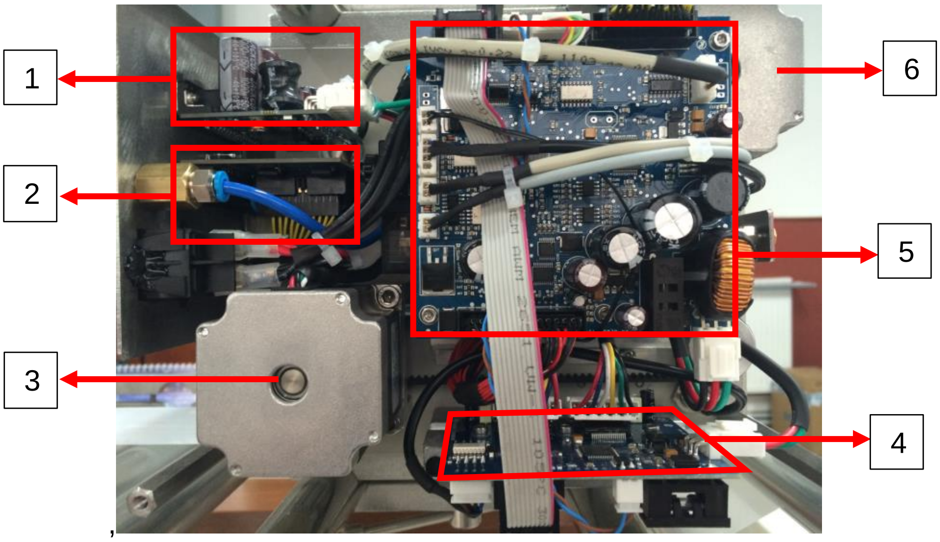


Figure 52. Printer Electronic Boards Positions

No	Designation	Spare Part Number
1	SCREEN COMMUNICATION BOARD (POWER BOARD)	200017-YI/70
2	CONTACT BOARD	200017-YO/70
3	HORİZONTAL MOTOR	200017-M2
4	MOTOR DRIVER	200017-MDB/70
5	MAINBOARD	200017-MB/70
6	RIBBON MOTOR	200017-M2

Table 31. Printer Electronic Boards Descriptions

4.1.1 MAINBOARD

The names and locations of the sockets on the mainboard are shown below;

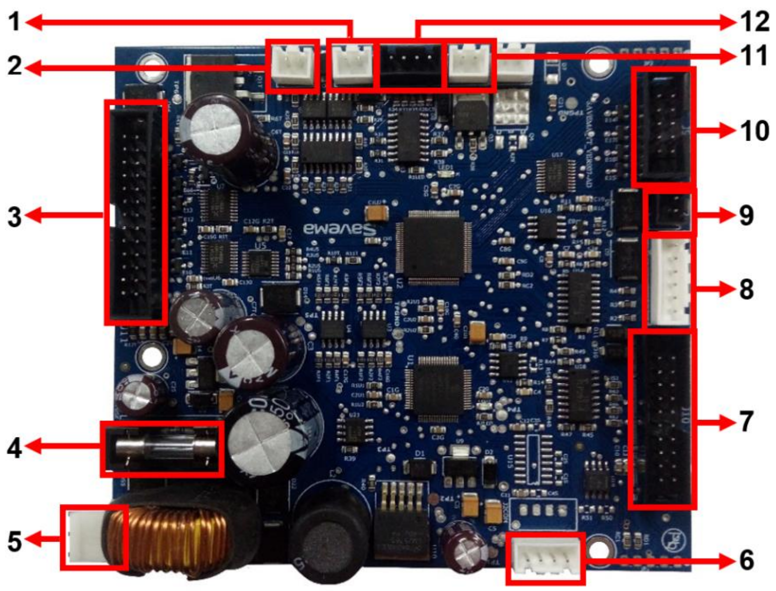


Figure 53. Connectors in Mainboard

No	Designation
1	STOP VALVE SOCKET
2	PRINT VALVE SOCKET
3	PRINTHEAD CABLE SOCKET
4	FUSE 2A
5	POWER INPUT SOCKET
6	SCREEN COMMUNICATION SOCKET
7	CONTACT BOARD SOCKET
8	CASSETTE AND RIBBON SENSORS SOCKET
9	MOTOR DRIVER CANBUS SOCKET
10	MOTOR DRIVER COMMUNICATION CABLE SOCKET
11	PRINTHEAD HEATER SOCKET
12	SWITCH SOCKET

Table 32. Connector Descriptions in Mainboard

4.1.2 MOTOR DRIVER BOARD

The names and locations of the sockets on the motor driver board are shown below;

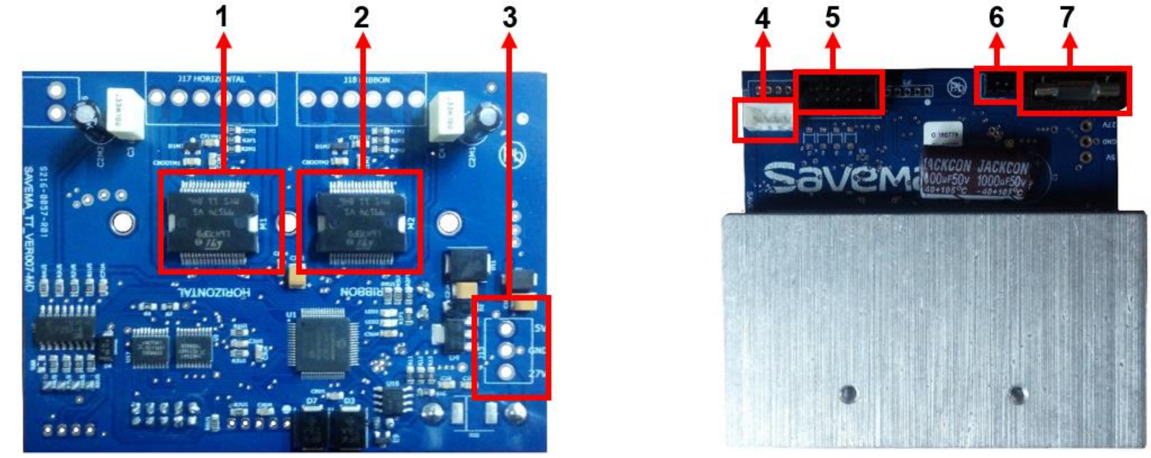


Figure 54. Connectors in Motor Driver Board (Left→Back Side, Right→ Front Side)

No	Designation	Spare Part Number
1	HORİZONTAL MOTOR DRIVER	-
2	RIBBON MOTOR DRIVER	-
3	POWER INPUT SOCKET	200017-PDP/70 (for cable)
4	HORIZONTAL LIMIT SENSOR SOCKET	200017-LS (for sensor)
5	MAINBOARD COMMUNICATION CABLE SOCKET	200017-MDC/70 (for cable)
6	CANBUS SOCKET	200017-MDCB/70 (for cable)
7	FUSE 3A	200017-F3

Table 33. Connector Descriptions in Motor Driver Board

4.1.3 SCREEN COMMUNICATION BOARD (POWER BOARD)

The names and locations of the sockets on the screen communication board are shown below;

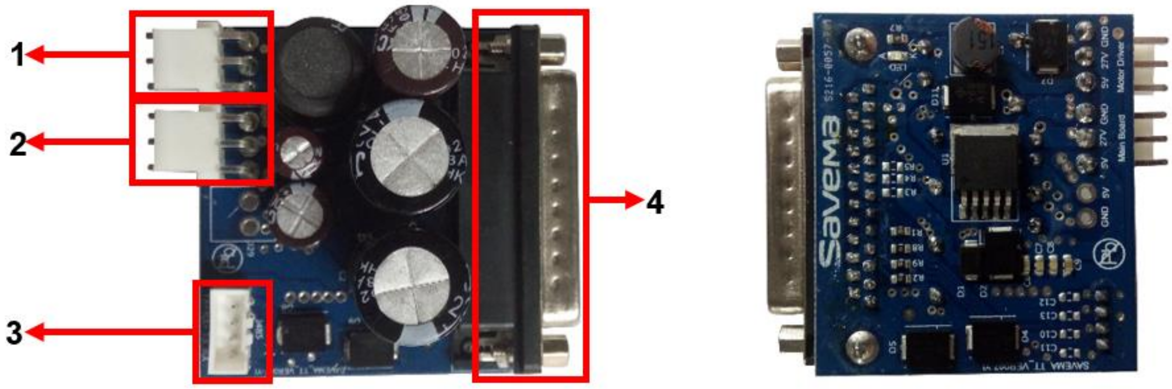


Figure 55. Connectors in Screen Communication Board (Left→Front Side, Right→ Back Side)

No	Designation	Spare Part Number
1	MOTOR DRIVER POWER OUTPUT SOCKET	200017-PDP/70 (for cable)
2	MAINBOARD POWER OUTPUT SOCKET	200017-PMP/70 (for cable)
3	UART CABLE SOCKET	200017-MPC/70 (for cable)
4	DB25 MALE SCREEN COMMUNICATION CABLE CONTACTOR	-

Table 34. Connector Descriptions in Screen Communıcation Board

4.1.4 CONTACT BOARD

The names and locations of the sockets on the contact board are shown below;

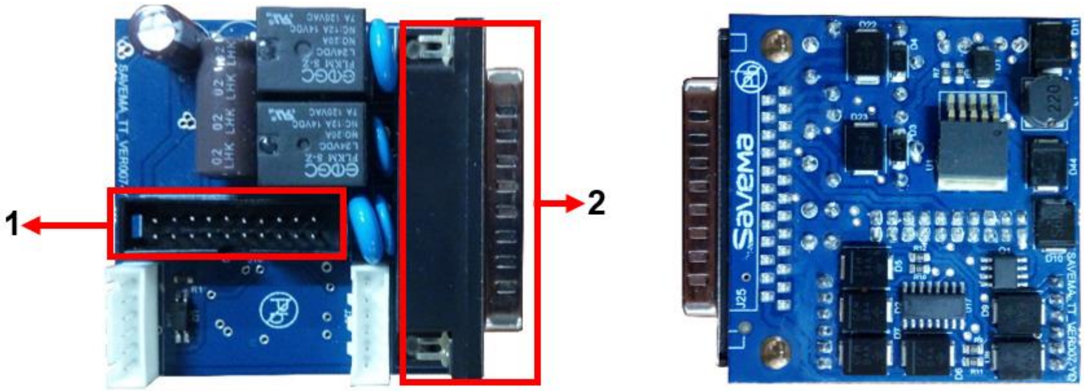


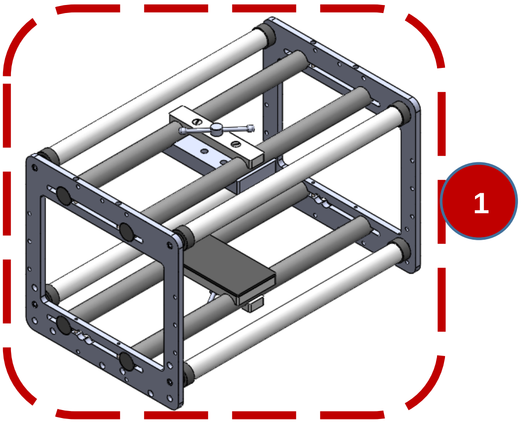
Figure 56. Connectors in Contact Board (Left→Front Side, Right→ Back Side)

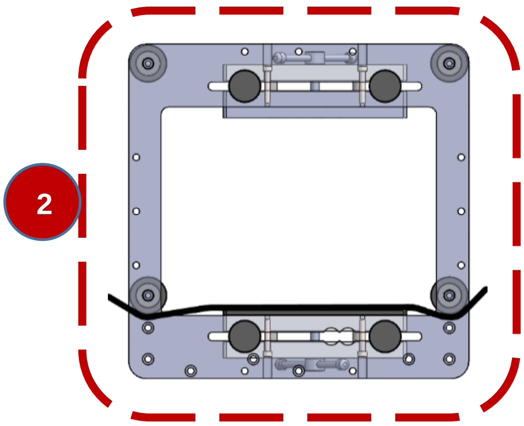
No	Designation	Spare Part Number
1	MAINBOARD CABLE SOCKET	200017-MCC/70 (for cable)
2	DB25 FEMALE CONTACT CABLE CONNECTOR	-

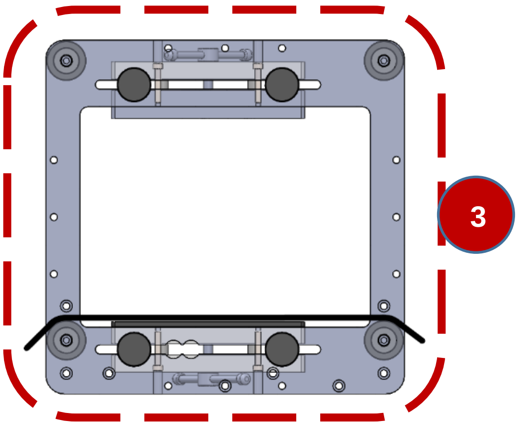
Table 35. Connector Descriptions in Contact Board

4.2 PACKAGE WINDING SYSTEM

There are 2 options for pack winding system like below;







No	Designation
1	EMPTY BRACKET
2	OPTION 1
3	OPTION 2

Figure 57. Correct Package Connection

4.3 RIBBON BREAK SYSTEM

The ribbon and cassette sensors in the machine are shown below;

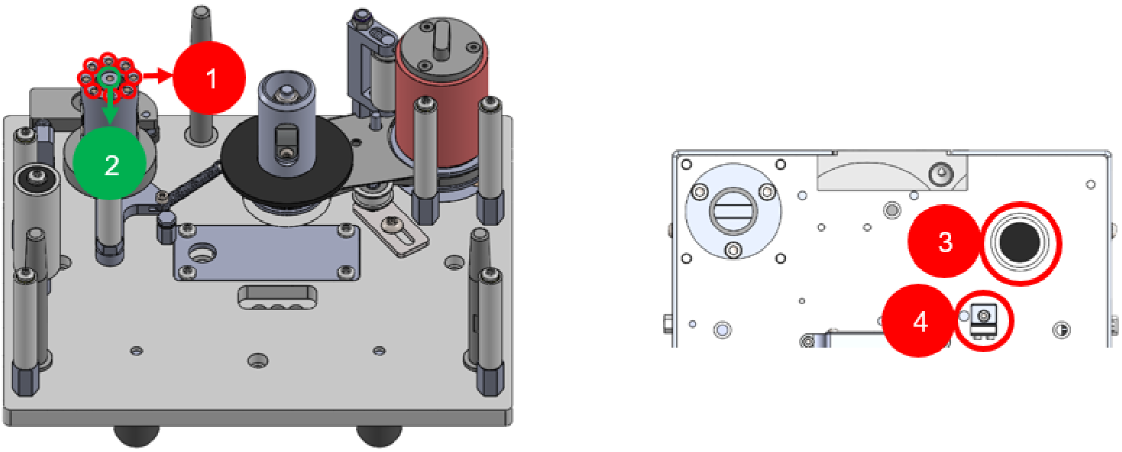


Figure 58. Cassette (Left) and Cassette Slot (Right)

No	Designation	Spare Part Number
1	RIBBON SENSOR TRANSMITTER	-
2	CASETTE SENSOR TRANSMITTER	-
3	RIBBON AND CASSETTE SENSOR RECEIVER	200017-CS
4	RIBBON BREAK SWITCH	200017-SWU

Table 36. Sensor Descriptions

4.4 THERMAL PRINT HEAD AND PRINTING SURFACE CLEANING

Important warning: Please turn off the power of the printer before cleaning the head.
After turning off the power,the head is cleaned with the help of and alcohol cotton pad. It takes a while for alcohol to disappear. Then the printer can be used again
Thermal print head can be cleaned with pure alcohol cotton like picture.

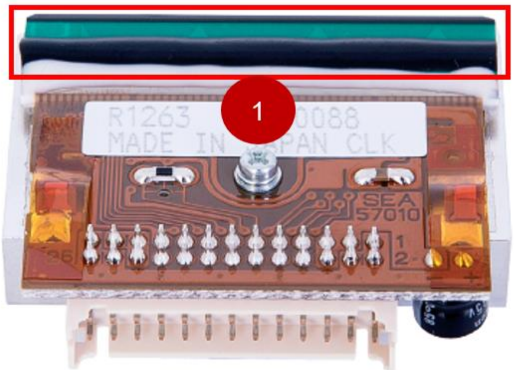


Figure 59. Thermal Print Head
Spare Part Number: 200017-TPH

No	Designation
1	CLEANING AREA

Table 37. Cleaning Area Description

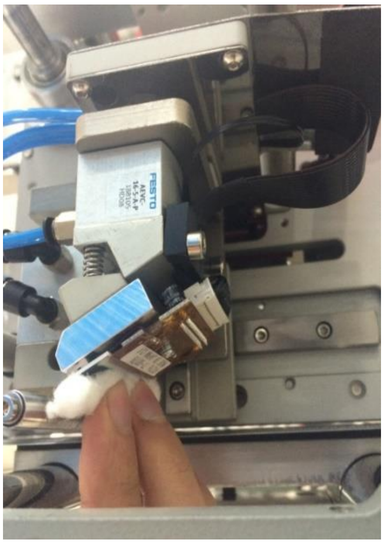


Figure 60. Cleaning of TPH

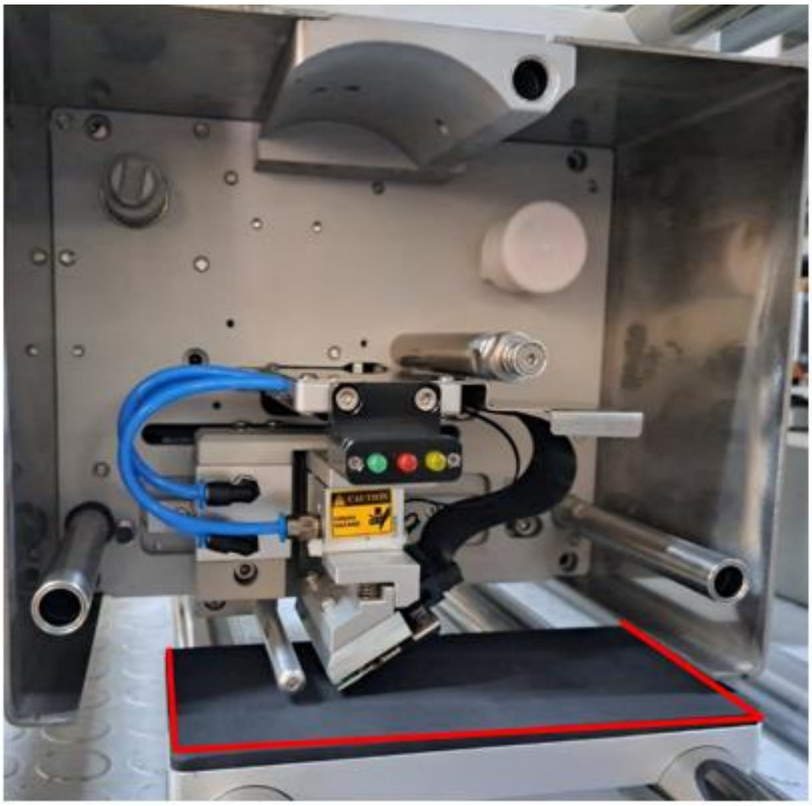


Figure 61 Printing Surface

Manufacturer's recommendation: If you clean the printing surface and head daily, you will extend the life of the machine.

4.5 HOME POSITION SENSOR (HORIZONTAL LIMIT SENSOR)

The horizontal limit sensor is used for the printer to find the home position. If the printer's head touches the limit sensor, the horizontal limit sensor in the control menu is shown in green
The horizontal limit sensor in the machine is shown below;

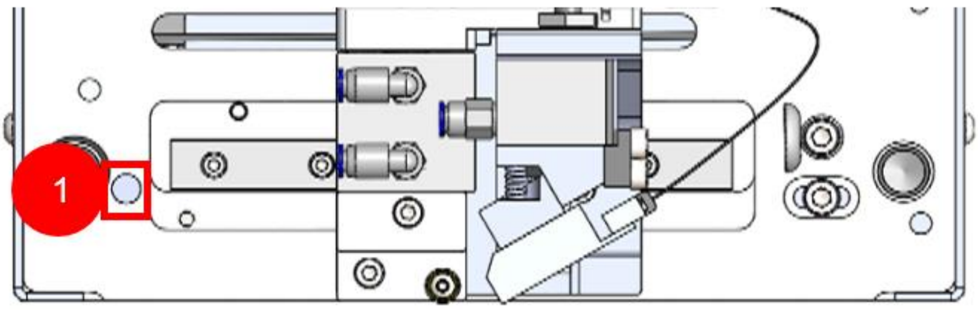


Figure 62. Horizontal Limit Sensor

No	Designation	Spare Part Number
1	HORIZONTAL LIMIT SENSOR POSITION	200017-LS

Table 38. Horizontal Limit Sensor Description

4.6 RECOMMENDED SPARE PARTS

No	Part Number	Recommended Spare Part Name	Compatibility for SVM 32*70 I Version 6[1]
1	PH	PHOTOCELL	✓
2	GK-32	POWER SUPPLY	X
3	200017-M2	RIBBON STEP MOTOR	X
4	200017-M2	HORIZONTAL STEP MOTOR	✓
5	200017-YI/70	POWER BOARD	X
6	200017-YO/70	CONTACT BOARD	X
7	200017-MB/70	MAINBOARD	X
8	200017-MDB/70	MOTOR DRIVER BOARD	X
9	200017-TCCB	TPH CABLE CONNECTION BOARD	✓
10	200017-TC24	TPH CABLE – 24 CM	✓
11	200017-TH2	PRINT HEAD HEATER	✓
12	200017-SWU	MICRO SWITCH	✓
13	200017-NFB	BLACK ON/OFF BUTTON	✓
14	200017-F3	FUSE 3 AMPER	✓
15	200017-F2	FUSE 2 AMPER	✓
16	200017-TPH	THERMAL PRINT HEAD	✓
17	200017-CS	CASSETTE SENSOR	✓
18	200017-LS	LIMIT SENSOR	✓
19	EM430-10	COM3 BOARD	X
20	C-DATAV7-43	CONNECTION CABLE FOR 4.3 INCH CONTROLLER	X
21	C-CNTV7	CONTACT CABLE FOR ALL PRINTERS	✓
22	C-SETV7-43	CONNECTION CABLE FOR 4.3 INCH CONTROLLER + CONTACT CABLE	X
23	GK-DB15	SECONDARY CABLE BETWEEN POWER SUPPLY AND PRINTER	X
24	200017-MPC/70	COMMUNICATION CABLE BETWEEN MAINBOARD AND POWER BOARD	✓
25	200017-MCC/70	COMMUNICATION CABLE BETWEEN MAINBOARD AND CONTACT BOARD	✓
26	200017-MDC/70	COMMUNICATION CABLE BETWEEN MAINBOARD AND MOTOR DRIVER BOARD	✓
27	200017-MDCB/70	CANBUS CABLE BETWEEN MAINBOARD AND MOTOR DRIVER BOARD	✓
28	200017-PMP/70	POWER CABLE BETWEEN MAINBOARD AND POWER BOARD	✓
29	200017-PDP/70	POWER CABLE BETWEEN MOTOR DRIVER BOARD AND POWER BOARD	✓

Table 39. Recommended Spare Part List

> **NOTE:** Differences between service manual and purchased device may be due to version update. In this case, please contact the manufacturer.

[1] If the table is filled with a "✓", this part can be used on the specified machine. If it is filled with an "X", using it on the specified machine may damage the machine. Our company is not responsible for this damage.